Exercise Solutions

Chapter 1

Exercise 1.1 According to Definition 1.4, we need to check the following five properties of Brownian motion:

(i) starts at 0 at time 0,

(ii) independence of increments,

(iii) almost sure continuity of trajectories,

(iv) stationarity of the increments,

(v) Gaussianity of increments.

Checking conditions (i) to (iv) does not pose any particular problem since the time changes $t \mapsto c + t$ and $t \mapsto t/c^2$ are deterministic and continuous.

a) Let $X_t := B_{c+t} - B_t$, $t \in \mathbb{R}_+$. For any finite sequence of times $t_0 < t_1 < \cdots < t_n$, the sequence

$$\begin{aligned}&(X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \ldots, X_{t_n} - X_{t_{n-1}})\\ &\quad = (B_{c+t_1} - B_{c+t_0}, B_{c+t_2} - B_{c+t_1}, \ldots, B_{c+t_n} - B_{c+t_{n-1}})\end{aligned}$$

is made of independent random variables. Concerning (v), $X_t - X_s = B_{c+t} - B_{c+s}$ is normally distributed with mean zero and variance $t + c - (c + s) = t - s$ for any $0 \leqslant s < t$.

b) Let $X_t := cB_{t/c^2}-$, $t \in \mathbb{R}_+$. For any finite sequence of times $t_0 < t_1 < \cdots < t_n$, the sequence

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$$(X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \ldots, X_{t_n} - X_{t_{n-1}})$$
$$= (cB_{t_1/c^2} - cB_{t_0/c^2}, cB_{t_2/c^2} - cB_{t_1/c^2}, \ldots, cB_{t_n/c^2} - cB_{t_{n-1}/c^2})$$

is made of independent random variables. Concerning (v), $X_t - X_s = cB_{t/c^2} - cB_{s/c^2}$ is normally distributed with mean zero and variance $c^2(t/c^2 - s/c^2) = t - s$ for any $0 \leqslant s < t$, since

$$\mathrm{Var}\left[cB_{t/c^2}\right] = c^2\mathrm{Var}\left[B_{t/c^2}\right] = c^2 t/c^2 = t, \qquad t \geqslant 0.$$

Exercise 1.2 The solution to (1.18) is given by

$$S_t = S_0 \mathrm{e}^{\sigma B_t + (\mu - \sigma^2/2)t}, \qquad t \geqslant 0,$$

see the proof of Proposition 1.8 for details. The next code can be used to generate Figure 1.24.

```
N=1000;t=0:N;dt=1./N;nsim=10;sigma=.6;mu=.001;Z=c(rnorm(n=N,sd=sqrt(dt)));
plot(t*dt,exp(mu*t),xlab="time",ylab="Geometric Brownian motion",type="l",ylim=
  c(0,4),col=1,lwd=3)
lines(t*dt,exp(sigma*c(0,cumsum(Z))+mu*t-sigma*sigma*t*dt/2),xlab="time",type="l",
  ylim=c(0,4),col= 4)
```

Listing S.1: R code - Geometric Brownian simulation.

```
import numpy as np,matplotlib.pyplot as plt;N=1000;dt=1/N;t=np.linspace(0,1,N+1);sigma=.6
mu=1.0;Z=np.random.normal(0,np.sqrt(dt),N); W=np.concatenate(([0],np.cumsum(Z)))
plt.plot(t,np.exp(mu*t), 'k', lw=1); plt.plot(t,np.exp(sigma*W+mu*t-.5*sigma**2*t), 'b', lw=1)
plt.ylim(0,4); plt.xlabel('time'); plt.ylabel('Geometric Brownian motion'); plt.show()
```

Listing S.2: Python code - Geometric Brownian simulation.

Exercise 1.3

a) Those quantities can be computed from the expression of S_t^n as a function of the $\mathcal{N}(0,t)$ random variable B_t for $n \geqslant 1$. Namely, we have

$$\begin{aligned}
\mathbb{E}[S_t^n] &= \mathbb{E}\left[S_0^n \mathrm{e}^{n\sigma B_t - n\sigma^2 t/2 + nrt}\right] \\
&= S_0^n \mathrm{e}^{-n\sigma^2 t/2 + nrt} \mathbb{E}\left[\mathrm{e}^{n\sigma B_t}\right] \\
&= S_0^n \mathrm{e}^{-n\sigma^2 t/2 + nrt + n^2\sigma^2 t/2} \\
&= S_0^n \mathrm{e}^{nrt + (n-1)n\sigma^2 t/2},
\end{aligned}$$

where we used the Gaussian moment generating function (MGF) formula (A.41), *i.e.*

$$\mathbb{E}\left[\mathrm{e}^{n\sigma B_t}\right] = \mathrm{e}^{n^2\sigma^2 t/2}$$

for the normal random variable $B_t \simeq \mathcal{N}(0,t)$, $t > 0$.

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b) By the result of Question (a), we have $\mathbb{E}[S_t] = S_0 e^{rt}$ and

$$\begin{aligned}\mathbb{E}[S_t^2] &= \mathbb{E}\big[S_0^2 e^{2\sigma B_t - \sigma^2 t + 2rt}\big] \\ &= S_0^2 e^{-\sigma^2 t + 2rt}\mathbb{E}\big[e^{2\sigma B_t}\big] \\ &= S_0^2 e^{\sigma^2 t + 2rt}, \qquad t \geqslant 0.\end{aligned}$$

Exercise 1.4 From the solution of Exercise 1.3, we have

$$\mathbb{E}\big[S_t^{(i)}\big] = S_0^{(i)} e^{\mu t}, \qquad t \in [0,T], \quad i = 1,2,$$

and

$$\begin{aligned}\mathrm{Var}\big[S_t^{(i)}\big] &= \mathbb{E}\big[\big(S_t^{(i)}\big)^2\big] - \big(\mathbb{E}\big[S_t^{(i)}\big]\big) \\ &= \big(S_0^{(i)}\big)^2 e^{2\mu t + \sigma_i^2 t} - \big(S_0^{(i)}\big)^2 e^{2\mu t} \\ &= \big(S_0^{(i)}\big)^2 e^{2\mu t}\big(e^{\sigma_i^2 t} - 1\big), \quad t \in [0,T], \quad i = 1,2.\end{aligned}$$

Hence, we have

$$\mathrm{Var}\big[S_t^{(2)} - S_t^{(1)}\big] = \mathrm{Var}\big[S_t^{(1)}\big] + \mathrm{Var}\big[S_t^{(2)}\big] - 2\,\mathrm{Cov}\big(S_t^{(1)}, S_t^{(2)}\big)$$

with

$$\begin{aligned}\mathbb{E}\big[S_t^{(1)} S_t^{(2)}\big] &= \mathbb{E}\big[S_0^{(1)} S_0^{(2)} e^{2\mu t + \sigma_1 W_t^{(1)} - \sigma_1^2 t/2 + \sigma_2 W_t^{(2)} - \sigma_2^2 t/2}\big] \\ &= S_0^{(1)} S_0^{(2)} e^{2\mu t - \sigma_1^2 t/2 - \sigma_2^2 t/2}\mathbb{E}\big[e^{\sigma_1 W_t^{(1)} + \sigma_2 W_t^{(2)}}\big] \\ &= S_0^{(1)} S_0^{(2)} e^{2\mu t - \sigma_1^2 t/2 - \sigma_2^2 t/2} \exp\left(\frac{1}{2}\mathbb{E}\big[\big(\sigma_1 W_t^{(1)} + \sigma_2 W_t^{(2)}\big)^2\big]\right),\end{aligned}$$

with

$$\begin{aligned}\mathbb{E}\big[\big(\sigma_1 W_t^{(1)} + \sigma_2 W_t^{(2)}\big)^2\big] &= \mathbb{E}\big[\big(\sigma_1 W_t^{(1)}\big)^2\big] + 2\mathbb{E}\big[\sigma_1 W_t^{(1)} \sigma_2 W_t^{(2)}\big] + \mathbb{E}\big[\big(\sigma_2 W_t^{(2)}\big)^2\big] \\ &= \sigma_1^2 t + 2\rho\sigma_1\sigma_2 t + \sigma_2^2 t,\end{aligned}$$

hence

$$\mathbb{E}\big[S_t^{(1)} S_t^{(2)}\big] = S_0^{(1)} S_0^{(2)} e^{2\mu t + \rho\sigma_1\sigma_2 t},$$

and

$$\mathrm{Cov}\big(S_t^{(1)}, S_t^{(2)}\big) = \mathbb{E}\big[S_t^{(1)} S_t^{(2)}\big] - \mathbb{E}\big[S_t^{(1)}\big]\mathbb{E}\big[S_t^{(2)}\big] = S_0^{(1)} S_0^{(2)} e^{2\mu t}(e^{\rho\sigma_1\sigma_2 t} - 1),$$

and therefore

$$\begin{aligned}&\mathrm{Var}\left[S_t^{(2)}-S_t^{(1)}\right]\\&=\left(S_0^{(1)}\right)^2\mathrm{e}^{2\mu t}(\mathrm{e}^{\sigma_1^2 t}-1)+\left(S_0^{(2)}\right)^2\mathrm{e}^{2\mu t}(\mathrm{e}^{\sigma_2^2 t}-1)-2S_0^{(1)}S_0^{(2)}\mathrm{e}^{2\mu t}(\mathrm{e}^{\rho\sigma_1\sigma_2 t}-1)\\&=\mathrm{e}^{2\mu t}\big(\left(S_0^{(1)}\right)^2\mathrm{e}^{\sigma_1^2 t}+\left(S_0^{(2)}\right)^2\mathrm{e}^{\sigma_2^2 t}-2S_0^{(1)}S_0^{(2)}\mathrm{e}^{\rho\sigma_1\sigma_2 t}-\left(S_0^{(2)}-S_0^{(1)}\right)^2\big).\end{aligned}$$

Exercise 1.5 We have

$$\mathbb{E}\left[S_0\mathrm{e}^{\sigma B_t+\mu t-\sigma^2 t/2}\right]=S_0\mathrm{e}^{\mu t-\sigma^2 t/2}\mathbb{E}\left[\mathrm{e}^{\sigma B_t}\right]=S_0\mathrm{e}^{\mu t-\sigma^2 t/2}\mathrm{e}^{\sigma^2 t/2}=S_0\mathrm{e}^{\mu t}$$

and

$$\mathbb{E}[\log S_t]=\mathbb{E}\left[\log S_0+\sigma B_t+\mu t-\frac{\sigma^2 t}{2}\right]=(\log S_0)+\mu t-\frac{\sigma^2 t}{2},$$

hence

$$\mathrm{Theil}_t=\log\mathbb{E}[S_t]-\mathbb{E}[\log S_t]=\log S_0+\mu t-\left((\log S_0)+\mu t-\frac{\sigma^2 t}{2}\right)=\frac{\sigma^2 t}{2}.$$

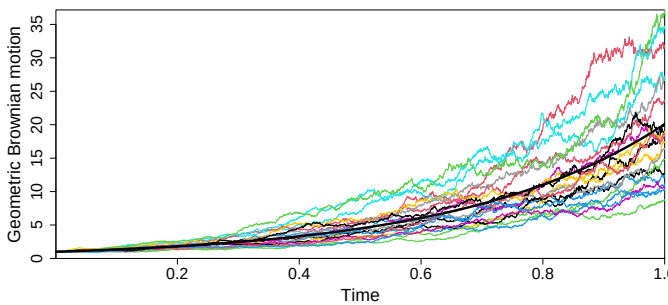


Fig. S.1: Twenty sample paths of geometric Brownian motion $(S_t)_{t\in\mathbb{R}_+}$.

Exercise 1.6 According to (1.13), taking $t=$ one year, we have

$$\mathbb{E}\left[\frac{S_t}{S_0}\right]=\mathrm{e}^{\mu t}=1+0.0949535=1.0949535$$

and

$$\mathrm{Var}\left[\frac{S_t}{S_0}\right]=\mathrm{e}^{2\mu t}\left(\mathrm{e}^{\sigma^2 t}-1\right)=0.1663085^2=0.02765852,$$

hence

$$\mu = \log 1.0949535 = 0.0907119$$

and

$$\begin{aligned}\sigma &= \sqrt{\log\left(1 + \mathrm{e}^{-2\mu t} \times 0.02765852\right)} \\ &= \sqrt{\log\left(1 + 0.02765852/1.0949535^2\right)} \\ &= 15.10212\%.\end{aligned}$$

Exercise 1.7

a) We have

$$\begin{aligned}dC_t &= dV_t - dA_t \\ &= \xi_t dS_t + \eta_t dA_t - dA_t \\ &= \beta \mathbf{1}_{\{V_t > A_t\}} \frac{V_t - A_t}{S_t} dS_t + \mathbf{1}_{\{V_t > A_t\}} \frac{V_t - (V_t - A_t)\beta}{A_t} dA_t \\ &\quad + \mathbf{1}_{\{V_t \leqslant A_t\}} \frac{V_t}{A_t} dA_t - dA_t \\ &= \beta \mathbf{1}_{\{V_t > A_t\}} \frac{V_t - A_t}{S_t} dS_t + \frac{V_t}{A_t} dA_t - \beta \mathbf{1}_{\{V_t > A_t\}} \frac{V_t - A_t}{A_t} dA_t - dA_t \\ &= \beta \mathbf{1}_{\{V_t > A_t\}} \frac{C_t}{S_t} dS_t + \frac{V_t - A_t}{A_t} dA_t - \beta \mathbf{1}_{\{V_t > A_t\}} \frac{C_t}{A_t} dA_t \\ &= \beta \frac{C_t}{S_t} dS_t + \frac{C_t}{A_t} dA_t - \beta \frac{C_t}{A_t} dA_t \\ &= \beta C_t(\mu dt + \sigma dB_t) + r C_t dt - r\beta C_t dt,\end{aligned}$$

hence

$$\frac{dC_t}{C_t} = ((\mu - r)\beta + r)dt + \beta\sigma dB_t.$$

b) We have

$$\begin{aligned}V_t &= A_t + C_t \\ &= A_0 e^{rt} + (V_0 - A_0) e^{((\mu - r)\beta + r)t + \beta\sigma B_t - \beta^2\sigma^2 t/2}.\end{aligned}$$

```r
N=1000; t=0:N; dt=1/N; tt=t*dt; sigma=.5; mu=1; beta=2.5; r=1; A0=1; C0=.1; S0=A0+C0
dB=rnorm(N,sd=sqrt(dt)); B=c(0,cumsum(dB)); par(mfrow=c(2,1), mai=c(.6,.5,.45,.2))
plot(tt,exp(r*tt),type="l",ylim=c(.8,3),xaxs="i",yaxs="i",xlab="t",ylab="",lwd=2,
 cex.axis=1.6,cex.lab=1.6)
A=A0*exp(r*tt); S=S0*exp(sigma*B + mu*tt - .5*sigma^2*tt)
C=C0*exp(beta*sigma*B + (beta*mu+(1-beta)*r)*tt - .5*beta^2*sigma^2*tt)
lines(tt,S,col=4,lwd=2); lines(tt,A+C,col=2,lwd=2); xi=beta*C; eta=(C+A-xi*S)/A
legend("topleft",c("Geometric Brownian motion","CPPI"),col=c(4,2),lty=1:2,cex=1.6)
plot(tt,xi,type="l",xaxs="i",yaxs="i",ylim=range(xi,eta),
 xlab="t",ylab="",lwd=2,cex.axis=1.6,cex.lab=1.6)
lines(tt,eta,col=6,lwd=2);
 legend("center",c(expression(xi),expression(eta)),col=c(1,6),lty=1:2,cex=1.6)
```

Listing S.3: R code - CPPI simulation.

```python
import numpy as np,matplotlib.pyplot as plt; N=1000; t=np.arange(N+1); dt=1/N; tt=t*dt
sigma=.5;mu=1;beta=2.5;r=1;A0=1;C0=.1;S0=A0+C0;
 dB=np.r_[0,np.random.normal(0,np.sqrt(dt),N)]
B=np.cumsum(dB); A=A0*np.exp(r*tt); S=S0*np.exp(sigma*B + mu*tt - .5*sigma**2*tt)
C=C0*np.exp(beta*sigma*B + (beta*mu+(1-beta)*r)*tt - .5*beta**2*sigma**2*tt)
fig,ax=plt.subplots(2,1,figsize=(10,8),constrained_layout=True)
ax[0].plot(tt,np.exp(r*tt),'k-',lw=2,label='Exp(r t)'); ax[0].plot(tt,S,'b-',lw=2,label='GBM')
ax[0].plot(tt,A+C,'r-',lw=2,label='CPPI'); ax[0].set(xlim=(0,1),ylim=(.8,3),xlabel='t');
ax[0].legend(fontsize=12); xi=beta*C;
 eta=(C+A-xi*S)/A;ax[1].plot(tt,xi,'k-',lw=2,label=r'$\xi$')
ax[1].plot(tt,eta,'m-',lw=2,label=r'$\eta$'); ax[1].legend(fontsize=12);
ax[1].set(xlim=(0,1),ylim=(min(xi.min(),eta.min()),max(xi.max(),eta.max())),xlabel='t')
plt.show()
```

Listing S.4: Python code - CPPI simulation.

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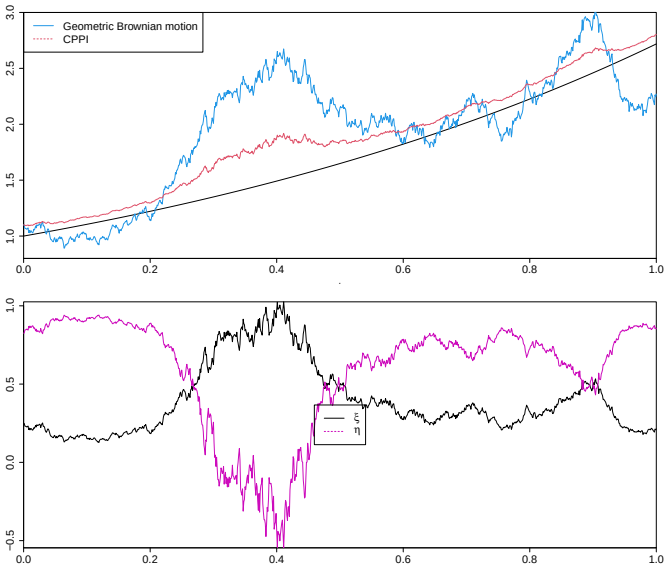


Fig. S.2: CPPI value and allocation.

Note that the mean return of C_t is $(\mu - r)\beta + r$, i.e. which is above r when $\mu > r$ and below r when $\mu < r$.

When $\mu = r$,

$$C_t = C_0 e^{rt + \beta\sigma B_t - \beta^2\sigma^2 t/2}$$

represents the value of a leveraged investment fund such as ProShares Ultra S&P500 or ProShares UltraShort S&P500, which are seeking daily investment results, before fees and expenses, that correspond to β times (βx) the daily performance of the S&P500,® with respectively $\beta = 2$ for ProShares Ultra and $\beta = -2$ for ProShares UltraShort. Here, *leveraging* with a factor $\beta : 1$ aims at multiplying the potential return of an investment by a factor β.

Note that $\mathbb{E}[C_t] = C_0 e^{rt}$ according to absence of arbitrage. On the other hand, we have

$$\mathrm{Var}[C_t] = C_0^2 e^{2rt}(e^{\beta^2\sigma^2 t} - 1), \quad t \geqslant 0,$$

which increases with $|\beta|$.

Exercise 1.8

a) From Table S.1 and the interest value \$40.83, we infer that ACBC bank used linear compounding of the form $1 + rt/365$.

Duration (days)	Linear compounding	Exponential compounding
182	\$40.39	\$40.66
183	\$40.61	\$40.89
184	\$40.83	\$41.11
185	\$41.05	\$41.34

Table S.1: Fixed deposit returns.

b) According to Table S.1, the investment duration was $t = 184$ days.

Remark: The actual investment dates were from March 11 to September 11, i.e.

$$31 - 11 + 30 + 31 + 30 + 31 + 31 + 11 = 184.$$

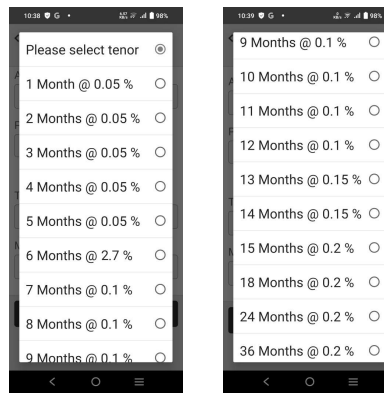


Fig. S.3: Phone screenshots.

Exercise 1.9

a) We have

$$\begin{aligned}\frac{dV(t)}{V(t)} &= \frac{dX_1(t)}{V(t)} + \cdots + \frac{dX_n(t)}{V(t)} \\ &= \pi_1(t)\frac{dX_1(t)}{X_1(t)} + \cdots + \pi_n(t)\frac{dX_n(t)}{X_n(t)} \\ &= \pi_1(t)\mu_1(t)dt + \cdots + \pi_n(t)\mu_n(t)dt\end{aligned}$$

$$+\pi_1(t)\sum_{k=1}^{n}\sigma_{1,k}(t)dB_k(t)+\cdots+\pi_n(t)\sum_{k=1}^{n}\sigma_{n,k}(t)dB_k(t)$$
$$=\bar{\mu}(t)dt+\sum_{k=1}^{n}\bar{\sigma}_k(t)dB_k(t),$$

where

$$\bar{\mu}(t)=\pi_1(t)\mu_1(t)+\cdots+\pi_n(t)\mu_n(t)$$

and

$$\bar{\sigma}_k(t)=\pi_1(t)\sigma_{1,k}(t)+\cdots+\pi_n(t)\sigma_{n,k}(t).$$

Remark: Note that from

$$\frac{dX_i(t)}{X_i(t)}=\mu_i(t)dt+\sum_{k=1}^{n}\sigma_{i,k}(t)dB_k(t),$$

we have

$$d\log X_i(t)=\frac{dX_i(t)}{X_i(t)}-\frac{(dX_i(t))^2}{2X_i^2(t)}$$
$$=\mu_i(t)dt+\sum_{k=1}^{n}\sigma_{i,k}(t)dB_k(t)-\frac{1}{2}\sum_{k=1}^{n}\sigma_{i,k}^2(t)dt,$$

hence the relation

$$\gamma_i(t)=\mu_i(t)-\frac{1}{2}\sum_{k=1}^{n}\sigma_{i,k}^2(t),\quad i=1,\ldots,n.$$

b) We have

$$d\log\pi_i(t)=d\log X_i(t)-d\log X(t)$$
$$=\gamma_i(t)dt+\sum_{k=1}^{n}\sigma_{i,k}(t)dB_k(t)-\bar{\gamma}(t)dt-\sum_{k=1}^{n}\bar{\sigma}_k(t)dB_k(t),$$
$$=(\gamma_i(t)-\bar{\gamma}(t))dt+\sum_{k=1}^{n}(\sigma_{i,k}(t)-\bar{\sigma}_k(t))dB_k(t).$$

Remark: Note that from

$$\frac{dV(t)}{V(t)}=\bar{\mu}(t)dt+\sum_{k=1}^{n}\bar{\sigma}_k(t)dB_k(t),$$

we have

$$d\log V(t) = \frac{dV(t)}{V(t)} - \frac{(dV(t))^2}{2V^2(t)}$$
$$= \bar{\mu}(t)dt + \sum_{k=1}^{n} \bar{\sigma}_k(t)dB_k(t) - \frac{1}{2}\sum_{k=1}^{n} \bar{\sigma}_k^2(t)dt,$$

hence

$$\bar{\gamma}(t) = \bar{\mu}(t) - \frac{1}{2}\sum_{k=1}^{n} \bar{\sigma}_k^2(t).$$

Chapter 2

Exercise 2.1

a) i) By calculation (expected answer). We have

$$\begin{aligned}\rho(0) &= \mathrm{Cov}(X_n, X_n)\\ &= \mathrm{Var}[X_n]\\ &= \mathbb{E}[X_n^2]\\ &= \mathbb{E}[(Z_n - aZ_{n-1})^2]\\ &= \mathbb{E}[Z_n^2 - 2aZ_{n-1}Z_n + a^2Z_{n-1}^2]\\ &= \mathbb{E}[Z_n^2] - 2a\mathbb{E}[Z_{n-1}Z_n] + a^2\mathbb{E}[Z_{n-1}^2]\\ &= 1 - 2a\mathbb{E}[Z_{n-1}]\mathbb{E}[Z_n] + a^2\\ &= 1 + a^2,\end{aligned}$$

and

$$\begin{aligned}\rho(1) &= \mathrm{Cov}(X_n, X_{n+1})\\ &= \mathrm{Cov}(Z_n + aZ_{n-1}, Z_{n+1} + aZ_n)\\ &= \mathrm{Cov}(Z_n, Z_{n+1}) + a\,\mathrm{Cov}(Z_{n-1}, Z_{n+1})\\ &\quad + a\,\mathrm{Cov}(Z_{n-1}, Z_n) + a^2\,\mathrm{Cov}(Z_{n-1}, Z_n)\\ &= \mathrm{Cov}(Z_n, Z_{n+1}) + a\,\mathrm{Cov}(Z_{n-1}, Z_{n+1})\\ &\quad + a\,\mathrm{Cov}(Z_n, Z_n) + a^2\,\mathrm{Cov}(Z_{n-1}, Z_n)\\ &= a\,\mathrm{Var}[Z_n]\\ &= a,\end{aligned}$$

and for $k \geqslant 2$,

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$$\begin{aligned}\rho(k) &= \mathrm{Cov}(X_n, X_{n+k}) \\ &= \mathrm{Cov}(Z_n + aZ_{n-1}, Z_{n+k} + aZ_{n+k-1}) \\ &= \mathrm{Cov}(Z_n, Z_{n+k}) + a\,\mathrm{Cov}(Z_{n-1}, Z_{n+k}) \\ &\quad + a\,\mathrm{Cov}(Z_n, Z_{n+k-1}) + a^2\,\mathrm{Cov}(Z_{n-1}, Z_{n+k-1}) \\ &= 0,\end{aligned}$$

since $n < n+k-1$. By a similar argument, we obtain $\rho(k) = \rho(-k)$ for $k \leqslant 0$.

ii) Confirmation by simulation with an MA(1) time series constructed by hand:

```r
library(zoo); N=10000;Zn<-zoo(rnorm(N,0,1))
Xn<-Zn+2*lag(Zn,-1, na.pad = TRUE);Xn<-Xn[-1]
k=0;cov(Xn[1:(length(Xn)-k)],lag(Xn,k))
```

Listing S.5: R code - MA(1) time series generation.

```python
import numpy as np, pandas as pd; N=10000;
Zn=np.random.normal(0,1,N); Xn=Zn[1:]+2*Zn[:-1]; k=0;
cov_result=np.cov(Xn[:-k] if k>0 else Xn, (np.roll(Xn,-k)[:-k] if k>0 else Xn))[0,1];
print(f"Covariance for k={k}: {cov_result}")
```

Listing S.6: Python code - MA(1) covariance estimation.

or with an MA(1) time series.

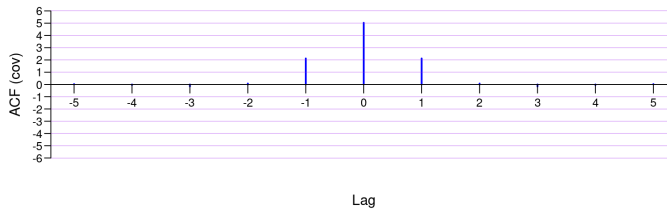
```r
n=2000;a=2; Xn<-arima.sim(model=list(ma=c(a)),n.start=100,n); x=seq(100,100+n-1)
plot(x,Xn,pch=19, ylab="X", xlab="n", main = 'MA(1) Samples',col='blue')
lines(x,Xn,col='blue'); Xn<-zoo(Xn)
k=1;cov(Xn[1:(length(Xn)-k)],lag(Xn,k))
```

Listing S.7: R code - MA(1) time series generation.

By computing the empirical autocovariance, we find the following.

```r
ccf(Xn,Xn,lag = 5, type="covariance", plot=T, lwd=2, col='blue', axes=FALSE,
  ylim=c(-1-a**2,1+a**2), main=""); axis(side = 1, at = seq(-5,5,1))
axis(side = 2, at = c(-1-a**2,-a,0,a,1+a**2), labels=c(expression(paste("-1-|a|"^"2")),
  "a",0,"a", expression(paste("1+|a|"^"2"))))
```

Listing S.8: R code - MA(1) covariance estimation.




```
import numpy as np, matplotlib.pyplot as plt, pandas as pd;
n=2000; a=2; e=np.random.normal(0,1,n+1);
Xn=pd.Series(e[1:]+a*e[:-1]); x=np.arange(100,100+n);
plt.plot(x,Xn,marker='o',ls='-',c='b',ms=2);
plt.xlabel("n"); plt.ylabel("X"); plt.title("MA(1) Samples");
plt.show(); k=1; print(np.cov(Xn[:-k],Xn.shift(k).dropna())[0,1])
```

Listing S.9: Python code - MA(1) time series generation.

```
import numpy as np, matplotlib.pyplot as plt
a=2;max_lag=5;Xc=Xn-np.mean(Xn); lags=np.arange(-max_lag,max_lag+1)
covs=[np.cov(Xc[:-lag] if lag>0 else Xc[-lag:],Xc[lag:] if lag>0 else Xc[:lag])[0,1] if
  lag!=0 else np.var(Xc) for lag in lags]
plt.bar(lags,covs,color='b',width=.8);plt.ylim(-1-a**2,1+a**2);plt.xticks(np.arange(-5,6))
plt.yticks([-1-a**2,-a,0,a,1+a**2],[r"$-1-|a|^2$","-a","0","a",r"$1+|a|^2$"])
plt.grid(True, ls='--', alpha=.5); plt.xlabel('Lag'); plt.ylabel('Covariance')
plt.title(''); plt.tight_layout(); plt.show()
```

Listing S.10: Python code - MA(1) covariance estimation.

b) We note that $\mathbb{E}[X_n] = 0$, $n \in \mathbb{Z}$, and in addition the autocovariance $\mathrm{Cov}(X_n, X_{n+k})$ depends only of $|k|$ and not on $n \in \mathbb{Z}$. Therefore, the time series $(X_n)_{n\in\mathbb{Z}}$ is *weakly stationary* by Definition 2.11. In addition, by Theorem 2.12 this time series is strictly stationary only if $|a| \neq 1$.

Exercise 2.2

a) We rewrite the equation defining $(X_n)_{n\geqslant 1}$ as

$$X_n = Z_n + LX_n = Z_n + \phi(L)X_n, \quad n \geqslant 1.$$

where L is the lag operator $LX_n = X_{n-1}$ and $\phi(L) = L$. Taking $\phi(z) := z$, by Theorem 2.12 we need to check whether the solutions of the equation $\phi(z) = 1$ lie on the complete unit circle. As $\phi(z) = 1$ admits the unique solution $z = 1$ which lies on the complete unit circle, we conclude that the AR(1) time series $(X_n)_{n\geqslant 1}$ is not weakly stationary.

b) As in part (a), we rewrite the AR(2) equation for $(Y_n)_{n\geqslant 1}$ as

$$Y_n = Z_n + 0.75 \times LY_n - 0.125 \times L^2 Y_n = Z_n + \phi(L)Y_n, \qquad n \geqslant 2.$$

with $\phi(L) = 0.75 \times L - 0.125 \times L^2$. The equation $\phi(z) = 1$ with $\phi(z) = 0.75z - 0.125z^2$ reads $z^2 - 6z + 8 = (z-2)(z-4) = 0$. This equation has two solutions $z = 2, 4$ which lie outside the complex unit circle, hence by Theorem 2.12 the AR(2) time series $(Y_n)_{n\geqslant 2}$ is weakly stationary.

Exercise 2.3

a) We have

$$\begin{aligned}\mathbb{E}[X_{n+1}] &= \mathbb{E}\big[Z_{n+1} + \alpha X_n\big] \\ &= \mathbb{E}[Z_{n+1}] + \alpha\mathbb{E}\big[X_n\big] \\ &= \alpha\mathbb{E}\big[X_n\big], \qquad n \geqslant 0,\end{aligned}$$

hence by induction we have $\mathbb{E}[X_n] = \alpha^n\mathbb{E}[X_0] = 0$ for all $n \geqslant 0$.

b) We have

$$\begin{aligned}\mathrm{Cov}(X_{n+k+1}, X_n) &= \mathbb{E}[X_{n+k+1}X_n] - \mathbb{E}[X_{n+k+1}]\mathbb{E}[X_n] \\ &= \mathbb{E}[X_{n+k+1}X_n] \\ &= \mathbb{E}[(Z_{n+k+1} + \alpha X_{n+k})X_n] \\ &= \mathbb{E}[Z_{n+k+1}X_n + \alpha X_{n+k}X_n] \\ &= \mathbb{E}[Z_{n+k+1}X_n] + \mathbb{E}[\alpha X_{n+k}X_n] \\ &= \mathbb{E}[Z_{n+k+1}]\mathbb{E}[X_n] + \alpha\mathbb{E}[X_{n+k}X_n] \\ &= \alpha\mathbb{E}[X_{n+k}X_n] \\ &= \alpha(\mathbb{E}[X_{n+k}X_n] - \mathbb{E}[X_{n+k}]\mathbb{E}[X_n]) \\ &= \alpha\,\mathrm{Cov}(X_{n+k}, X_n), \qquad n \geqslant 0,\end{aligned}$$

hence, since $\mathbb{E}[X_n] = 0$, $n \geqslant 0$, we find

$$\mathrm{Cov}(X_{n+k}, X_n) = \alpha^k\,\mathrm{Cov}(X_n, X_n) = \alpha^k\,\mathrm{Var}[X_n], \qquad k, n \geqslant 0.$$

c) We have

$$\begin{aligned}\mathrm{Var}[X_{n+1}] &= \mathbb{E}[X_{n+1}^2] \\ &= \mathbb{E}[(Z_{n+1} + \alpha X_n)^2] \\ &= \mathbb{E}[Z_{n+1}^2 + 2\alpha Z_{n+1}X_n + \alpha^2 X_n^2] \\ &= \mathbb{E}[Z_{n+1}^2] + 2\alpha\mathbb{E}[Z_{n+1}X_n] + \alpha^2\mathbb{E}[X_n^2] \\ &= 1 + 2\alpha\mathbb{E}[Z_{n+1}]\mathbb{E}[X_n] + \alpha^2\mathbb{E}[X_n^2] \\ &= 1 + \alpha^2\mathbb{E}[X_n^2] \\ &= 1 + \alpha^2\,\mathrm{Var}[X_n].\end{aligned}$$

By applying the above relation recursively and using the geometric series identity (13.52), we obtain

$$\begin{aligned}
\mathrm{Var}[X_n] &= 1+\alpha^2\,\mathrm{Var}[X_{n-1}] \\
&= 1+\alpha^2(1+\alpha^2\,\mathrm{Var}[X_{n-2}]) \\
&= 1+\alpha^2(1+\alpha^2(1+\alpha^2\,\mathrm{Var}[X_{n-2}])) \\
&= 1+\alpha^2+\cdots+\alpha^{2n} \\
&= \sum_{k=0}^{n}\alpha^{2k} \\
&= \begin{cases} \dfrac{1-\alpha^{2n+2}}{1-\alpha^2}, & \alpha\neq\pm 1, \\ \\ n+1, & \alpha=\pm 1, \qquad n\geqslant 0. \end{cases}
\end{aligned}$$

d) We check that the solution of $\phi(z) := \alpha z$ is $z = 1/\alpha$, hence by Theorem 2.12 there exists an AR(1) solution of (2.31) which is weakly stationary when $\alpha \neq \pm 1$. However, the present time series $(X_n)_{n\geqslant 0}$ started at $X_0 = 0$ is *not* weakly stationary because $\mathrm{Cov}(X_n, X_n) = \mathbb{E}[X_n^2]$ is not constant in $n \geqslant 0$.

Exercise 2.4

a) We have

$$\begin{aligned}
\mathrm{Var}[X_n] &= \mathrm{Var}[Z_{n-1} - Z_{n-2} + \alpha Z_{n-3}] \\
&= \mathrm{Var}[Z_{n-1}] + \mathrm{Var}[-Z_{n-2}] + \mathrm{Var}[\alpha Z_{n-3}] \\
&= \mathrm{Var}[Z_{n-1}] + \mathrm{Var}[Z_{n-2}] + \alpha^2\,\mathrm{Var}[Z_{n-3}] \\
&= 2+\alpha^2.
\end{aligned}$$

Next, since using the linearity relation

$$\mathrm{Cov}(X+Y,Z) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)$$

and the fact that $\mathrm{Cov}(X, Z) = 0$ when X and Z are independent random variables, we have

$$\begin{aligned}
\mathrm{Cov}(X_{n+1}, X_n) &= \mathrm{Cov}(Z_n - Z_{n-1} + \alpha Z_{n-2}, Z_{n-1} - Z_{n-2} + \alpha Z_{n-3}) \\
&= \mathrm{Cov}(Z_n - Z_{n-1} + \alpha Z_{n-2}, Z_{n-1}) \\
&\quad + \mathrm{Cov}(Z_n - Z_{n-1} + \alpha Z_{n-2}, -Z_{n-2}) \\
&\quad + \mathrm{Cov}(Z_n - Z_{n-1} + \alpha Z_{n-2}, \alpha Z_{n-3}) \\
&= \mathrm{Cov}(-Z_{n-1}, Z_{n-1}) + \mathrm{Cov}(\alpha Z_{n-2}, -Z_{n-2})
\end{aligned}$$

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$$\begin{aligned}&= -\operatorname{Cov}(Z_{n-1}, Z_{n-1}) - \alpha \operatorname{Cov}(Z_{n-2}, Z_{n-2})\\ &= -\alpha - 1,\end{aligned}$$

and

$$\begin{aligned}\operatorname{Cov}(X_{n+2}, X_n) &= \operatorname{Cov}(Z_{n+2} - Z_n + \alpha Z_{n-1}, Z_{n-1} - Z_{n-2} + \alpha Z_{n-3})\\ &= \operatorname{Cov}(Z_{n+2} - Z_n + \alpha Z_{n-1}, Z_{n-1})\\ &\quad + \operatorname{Cov}(Z_{n+2} - Z_n + \alpha Z_{n-1}, -Z_{n-2})\\ &\quad + \operatorname{Cov}(Z_{n+2} - Z_n + \alpha Z_{n-1}, \alpha Z_{n-3})\\ &= \operatorname{Cov}(\alpha Z_{n-1}, Z_{n-1})\\ &= \alpha \operatorname{Cov}(Z_{n-1}, Z_{n-1})\\ &= \alpha,\end{aligned}$$

and

$$\begin{aligned}\operatorname{Cov}(X_{n+k}, X_n) &= \operatorname{Cov}(Z_{n+k-1} - Z_{n+k-2} + \alpha Z_{n+k-3}, Z_{n-1} - Z_{n-2} + \alpha Z_{n-3})\\ &= \operatorname{Cov}(Z_{n+k-1} - Z_{n+k-2} + \alpha Z_{n+k-3}, Z_{n-1})\\ &\quad + \operatorname{Cov}(Z_{n+k-1} - Z_{n+k-2} + \alpha Z_{n+k-3}, -Z_{n-2})\\ &\quad + \operatorname{Cov}(Z_{n+k-1} - Z_{n+k-2} + \alpha Z_{n+k-3}, \alpha Z_{n-3})\\ &= 0\end{aligned}$$

for $k \geqslant 3$.

b) Since the white noise sequence $(Z_n)_{n\in\mathbb{Z}}$ is made of independent identically distributed random variables, we have the identity in distribution

$$X_n = Z_{n-1} - Z_{n-2} + \alpha Z_{n-3} \overset{d}{=} Z_n - Z_{n-1} + \alpha Z_{n-2}, \qquad n \geqslant 2,$$

which shows that $(X_n)_{n\geqslant 3}$ has the same distribution as an MA(2) time series the form

$$Y_n = Z_n + \beta_1 Z_{n-1} + \beta_2 Z_{n-2},$$

with $\beta_1 = -1$ and $\beta_1 = \alpha$.

Exercise 2.5

a) We have

$$\begin{aligned}\nabla X_n &= X_n - X_{n-1}\\ &= Z_n + \alpha_1 X_{n-1} - Z_{n-1} - \alpha_1 X_{n-2}\\ &= Z_n - Z_{n-1} + \alpha_1 \nabla X_{n-1}, \qquad n \geqslant 2,\end{aligned}$$

hence $(\nabla X_n)_{n\geqslant 2}$ forms an ARMA(1, 1) time series.

b) We have

$$\begin{aligned}\nabla^2 X_n &= \nabla X_n - \nabla X_{n-1}\\ &= X_n - X_{n-1} - (X_{n-1} - X_{n-2})\\ &= X_n - 2X_{n-1} + X_{n-2}\\ &= Z_n + \alpha_1 X_{n-1} - 2Z_{n-1} - 2\alpha_1 X_{n-2} + Z_{n-2} + \alpha_1 X_{n-3}\\ &= Z_n - 2Z_{n-1} + Z_{n-2} + \alpha_1 \nabla^2 X_{n-1}, \qquad n \geqslant 3,\end{aligned}$$

which forms an ARMA$(1,2)$ time series.

Exercise 2.6

a) We have

$$\begin{aligned}\frac{\partial}{\partial a}\sum_{k=1}^n \big(r_k^{(2)} - a - br_k^{(1)}\big)^2 &= -2\left(\sum_{k=1}^n \big(r_k^{(2)} - a - br_k^{(1)}\big)\right)\\ &= 2an - 2\sum_{k=1}^n r_k^{(2)} + 2b\sum_{k=1}^n r_k^{(1)},\end{aligned}$$

and

$$\begin{aligned}&\frac{\partial}{\partial b}\sum_{k=1}^n \big(r_k^{(2)} - a - br_k^{(1)}\big)^2 = 2\sum_{k=1}^n r_k^{(1)}\big(-a + r_k^{(2)} - br_k^{(1)}\big)\\ &= 2\sum_{k=1}^n r_k^{(1)}\left(r_k^{(2)} - br_k^{(1)} - \frac{1}{n}\sum_{l=1}^n \big(r_l^{(2)} - br_l^{(1)}\big)\right)\\ &= 2\sum_{k=1}^n r_k^{(1)} r_k^{(2)} - \frac{2}{n}\sum_{k,l=1}^n r_k^{(1)} r_l^{(2)} - 2b\left(\sum_{k=1}^n (r_k^{(1)})^2 - \frac{1}{n}\sum_{k,l=1}^n r_k^{(1)} r_l^{(1)}\right).\end{aligned}$$

b) In order to minimize the residual (2.32) over a and b we equate the above derivatives to zero, which yields the equations

$$\frac{\partial}{\partial a}\sum_{k=1}^n \big(r_k^{(2)} - a - br_k^{(1)}\big)^2_{|a=\widehat{a},\ b=\widehat{b}} = 2\widehat{a}n - 2\sum_{k=1}^n r_k^{(2)} + 2\widehat{b}\sum_{k=1}^n r_k^{(1)} = 0$$

and

$$\frac{\partial}{\partial b}\sum_{k=1}^n \big(r_k^{(2)} - a - br_k^{(1)}\big)^2_{|a=\widehat{a},\ b=\widehat{b}}$$

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$$= 2\sum_{k=1}^{n} r_k^{(1)} r_k^{(2)} - \frac{2}{n}\sum_{k,l=1}^{n} r_k^{(1)} r_l^{(2)} - 2\widehat{b}\left(\sum_{k=1}^{n}(r_k^{(1)})^2 - \frac{1}{n}\sum_{k,l=1}^{n} r_k^{(1)} r_l^{(1)}\right)$$
$$= 0.$$

This leads to estimators $\widehat{a}$, $\widehat{b}$ of the parameters a and b respectively as the empirical mean and covariance of $(r_k^{(1)})_{k=1,2,\ldots,n}$, *i.e.*

$$\begin{cases} \widehat{a} = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n}\left(r_k^{(2)} - \widehat{b} r_k^{(1)}\right), \\ \\ \text{and} \\ \\ \widehat{b} = \dfrac{\displaystyle\sum_{k=1}^{n} r_k^{(1)} r_k^{(2)} - \frac{1}{n}\sum_{k,l=0}^{n} r_k^{(1)} r_l^{(2)}}{\displaystyle\sum_{k=1}^{n}(r_k^{(1)})^2 - \frac{1}{n}\sum_{k,l=0}^{n} r_k^{(1)} r_l^{(1)}} = \dfrac{\displaystyle\sum_{k=1}^{n}\left(r_k^{(1)} - \frac{1}{n}\sum_{l=0}^{n} r_l^{(1)}\right)\left(r_k^{(2)} - \frac{1}{n}\sum_{l=0}^{n} r_l^{(2)}\right)}{\displaystyle\sum_{k=1}^{n}\left(r_k^{(1)} - \frac{1}{n}\sum_{k=1}^{n} r_k^{(1)}\right)^2}. \end{cases}$$

Exercise 2.7 Since the p-value = 0.02377 is lower than the 5% confidence level, we can reject the nonstationarity (null) hypothesis H_0 at that level.

Exercise 2.8

a) We consider the equation

$$\varphi(z) = \alpha_1 z + \alpha_2 z^2 = 1,$$

i.e.

$$\alpha_2 z^2 + \alpha_1 z - 1 = 0,$$

with solutions

$$z_\pm = \frac{-\alpha_1 \pm \sqrt{\alpha_1^2 + 4\alpha_2}}{2\alpha_2} = \frac{-a \pm \sqrt{a^2 + 8a^2}}{4a^2} = \frac{-a \pm 3a}{4a^2} = \begin{cases} \dfrac{1}{2a} \\ -\dfrac{1}{a}, \end{cases}$$

hence by Theorem 2.12 the time series $(X_n)_{n\geqslant 1}$ is stationary for $a \notin \{-1, -1/2, 1/2, 1\}$.

b) We have

$$\mathbb{E}[X_n] = \mathbb{E}[Z_n + \alpha_1 X_{n-1} + \alpha_2 X_{n-2}]$$

$$
\begin{aligned}
&= \mathbb{E}[Z_n] + \alpha_1 \mathbb{E}[X_{n-1}] + \alpha_2 \mathbb{E}[X_{n-2}] \\
&= \alpha_1 \mathbb{E}[X_{n-1}] + \alpha_2 \mathbb{E}[X_{n-2}] \\
&= \alpha_1 \mathbb{E}[X_n] + \alpha_2 \mathbb{E}[X_n],
\end{aligned}
$$

hence

$$
(1 - \alpha_1 - \alpha_2)\mathbb{E}[X_n] = 0,
$$

which implies $\mathbb{E}[X_n] = 0$, $n \in \mathbb{Z}$, since $1 - \alpha_1 - \alpha_2 \neq 0$.

c) We have

$$
\begin{aligned}
\mathrm{Cov}(X_n, Z_n) &= \mathrm{Cov}(Z_n + \alpha_1 X_{n-1} + \alpha_2 X_{n-2}, Z_n) \\
&= \mathrm{Cov}(Z_n, Z_n) + \alpha_1 \,\mathrm{Cov}(Z_n, X_{n-1}) + \alpha_2 \,\mathrm{Cov}(Z_n, X_{n-2}) \\
&= \mathrm{Cov}(Z_n, Z_n) \\
&= \sigma^2.
\end{aligned}
$$

d) We have

$$
\begin{aligned}
\mathrm{Cov}(X_{n+1}, X_n) &= \mathrm{Cov}(Z_{n+1} + \alpha_1 X_n + \alpha_2 X_{n-1}, X_n) \\
&= \mathrm{Cov}(Z_{n+1}, X_n) + \alpha_1 \,\mathrm{Cov}(X_n, X_n) + \alpha_2 \,\mathrm{Cov}(X_{n-1}, X_n), \\
&= 16\alpha_1 + \alpha_2 \,\mathrm{Cov}(X_{n-1}, X_n) \\
&= 4 + \frac{1}{2} \,\mathrm{Cov}(X_{n-1}, X_n),
\end{aligned}
$$

hence

$$
\mathrm{Cov}(X_{n+1}, X_n) = 8, \qquad n \in \mathbb{Z}.
$$

Chapter 3

Exercise 3.1

a) Since $Z_1 + Z_2 + \cdots + Z_n$ has the centered Gaussian $\mathcal{N}(0, n\sigma^2)$ distribution with variance $n\sigma^2$, we have

$$
\begin{aligned}
\mathbb{P}(Y \geqslant y) &= \sum_{n \geqslant 1} \mathbb{P}\left(\sum_{k=1}^{N} Z_k \geqslant y \;\middle|\; N = n \right) \mathbb{P}(N = n) \\
&= \sum_{n \geqslant 1} \left(1 - \mathbb{P}\left(\sum_{k=1}^{n} Z_k < y \;\middle|\; N = n \right) \right) \mathbb{P}(N = n) \\
&= \sum_{n \geqslant 1} \left(1 - \mathbb{P}\left(\sum_{k=1}^{n} Z_k \leqslant y \;\middle|\; N = n \right) \right) \mathbb{P}(N = n)
\end{aligned}
$$

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$$
\begin{aligned}
&= \sum_{n\geqslant 1}\left(1-\Phi\left(\frac{y}{\sqrt{n\sigma^2}}\right)\right)\mathbb{P}(N=n)\\
&= \mathrm{e}^{-\lambda}\sum_{n\geqslant 1}\frac{\lambda^n}{n!}\Phi\left(-\frac{y}{\sqrt{n\sigma^2}}\right), \qquad y>0.
\end{aligned}
$$

b) Since $\mathbb{E}[Z_k]=0$ for all $k\geqslant 1$, we have

$$
\begin{aligned}
\mathbb{E}[Y] &= \sum_{n\geqslant 1}\mathbb{E}\left[\sum_{k=1}^{N} Z_k \;\middle|\; N=n\right]\mathbb{P}(N=n)\\
&= \sum_{n\geqslant 1}\mathbb{E}\left[\sum_{k=1}^{n} Z_k \;\middle|\; N=n\right]\mathbb{P}(N=n)\\
&= \sum_{n\geqslant 1}\mathbb{P}(N=n)\sum_{k=1}^{n}\mathbb{E}[Z_k \mid N=n]\\
&= \sum_{n\geqslant 1}\mathbb{P}(N=n)\sum_{k=1}^{n}\mathbb{E}[Z_k]\\
&= 0,
\end{aligned}
$$

as in (3.12).

Exercise 3.2 By (3.17), we have

$$
\begin{aligned}
\Phi'(y) &= \frac{\lambda}{c}\Phi(y)-\frac{\lambda}{c}\int_0^y \Phi(y-z)dF(z)\\
&= \frac{\lambda}{c}\Phi(y)-\frac{\lambda}{\mu c}\int_0^y \Phi(y-z)\mathrm{e}^{-z/\mu}dz\\
&= \frac{\lambda}{c}\Phi(y)-\frac{\lambda}{\mu c}\int_0^y \Phi(z)\mathrm{e}^{-(y-z)/\mu}dz,
\end{aligned}
$$

hence the differential equation

$$
\begin{aligned}
\Phi''(y) &= \frac{\lambda}{c}\Phi'(y)-\frac{\lambda}{\mu c}\Phi(y)+\frac{\lambda}{\mu^2 c}\int_0^y \Phi(z)\mathrm{e}^{-(y-z)/\mu}dz\\
&= \frac{\lambda}{c}\Phi'(y)-\frac{\lambda}{\mu c}\Phi(y)+\frac{1}{\mu}\left(\frac{\lambda}{c}\Phi(y)-\Phi'(y)\right)\\
&= \left(\frac{\lambda}{c}-\frac{1}{\mu}\right)\Phi'(y),
\end{aligned}
$$

which can be solved as

$$\Phi(y) = 1 - \frac{\lambda\mu}{c} \mathrm{e}^{(\lambda/c - 1/\mu)y},$$

given the boundary conditions $\Phi(\infty) = 1$ and $\Phi(0) = 1 - \lambda\mu/c$, cf. (3.19). We conclude that

$$\Psi(y) = \frac{\lambda\mu}{c} \mathrm{e}^{(\lambda/c - 1/\mu)y}, \qquad y \geqslant 0,$$

provided that $c < \lambda\mu$.

Exercise 3.3

a) We have

$$\begin{aligned} \mathbb{E}[R_T] &= \mathbb{E}[R_0 + \mu T - CN_T] \\ &= \mathbb{E}[R_0] + \mathbb{E}[\mu T] - \mathbb{E}[CN_T] \\ &= R_0 + \mu T - C\mathbb{E}[N_T] \\ &= R_0 + \mu T - C\lambda T \\ &= R_0 + (\mu - \lambda C)T, \end{aligned}$$

and similarly

$$\mathrm{Var}[R_T] = \mathrm{Var}[\mu T - CN_T] = \mathrm{Var}[-CN_T] = C^2 \,\mathrm{Var}[N_T] = \lambda C^2 T.$$

b) We find

$$\begin{aligned} \mathbb{P}(R_T < 0) &= \mathbb{P}(R_0 + \mu T - CN_T < 0) \\ &= \mathbb{P}(N_T > (R_0 + \mu T)/C) \\ &= \sum_{k > (R_0 + \mu T)/C} \mathbb{P}(N_T = k) \\ &= \mathrm{e}^{-\lambda T} \sum_{k > (R_0 + \mu T)/C} \frac{(\lambda T)^k}{k!}. \end{aligned}$$

Exercise 3.4

a) By Proposition 3.7 we have

$$\mathbb{E}[S(T)] = \lambda T \mathbb{E}[Z] \quad \text{and} \quad \mathrm{Var}[S(T)] = \lambda T \mathbb{E}[Z^2].$$

b) We have

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$$\begin{aligned}\mathbb{P}(x+f(T)-S(T)<0) &\leqslant \frac{\mathrm{Var}[x+f(T)-S(T)]}{(\mathbb{E}[x+f(T)-S(T)])^2}\\ &= \frac{\mathrm{Var}[S(T)]}{(x+f(T)-\mathbb{E}[S(T)])^2}\\ &= \frac{\lambda T\mathbb{E}[Z_1^2]}{(x+f(T)-\lambda T\mathbb{E}[Z_1])^2}.\end{aligned}$$

Chapter 4

Exercise 4.1

a) Taking $(U,V)=(U,U)$, we have

$$\begin{aligned}\mathbb{P}(U\leqslant u \text{ and } V\leqslant v) &= \mathbb{P}(U\leqslant u \text{ and } U\leqslant v)\\ &= \mathbb{P}(U\leqslant \min(u,v))\\ &= \min(u,v)\\ &= \overline{C}(u,v), \qquad u,v\in[0,1].\end{aligned}$$

b) Taking $(U,V)=(U,1-U)$, we have

$$\begin{aligned}\mathbb{P}(U\leqslant u \text{ and } V\leqslant v) &= \mathbb{P}(U\leqslant u \text{ and } 1-U\leqslant v)\\ &= \mathbb{P}(U\leqslant u \text{ and } U\geqslant 1-v)\\ &= \mathbb{P}(1-v\leqslant U\leqslant u)\\ &= \mathbb{1}_{\{0\leqslant 1-v\leqslant u\leqslant 1\}}\mathbb{P}(1-v\leqslant U\leqslant u)\\ &= \mathbb{1}_{\{0\leqslant u+v-1\leqslant 1\}}(u-(1-v))\\ &= (u+v-1)^+,\end{aligned}$$

$u,v\in[0,1]$.

c) We have

$$C(u,v)=\mathbb{P}(U\leqslant u \text{ and } V\leqslant v)\leqslant \mathbb{P}(U\leqslant u \text{ and } V\geqslant 1)\leqslant \mathbb{P}(U\leqslant u)=u,$$

$u,v\in[0,1]$, and similarly we find $C(u,v)\leqslant \mathbb{P}(U\leqslant v)=v$ for all $u,v\in[0,1]$, which yields (4.9).

d) For fixed $v\in[0,1]$ we have

$$\begin{aligned}\frac{\partial C}{\partial u}(u,v) &= \lim_{\varepsilon\to 0}\frac{C(u+\varepsilon,v)-C(u,v)}{\varepsilon}\\ &= \lim_{\varepsilon\to 0}\frac{\mathbb{P}(U\leqslant u+\varepsilon \text{ and } V\leqslant v)-\mathbb{P}(U\leqslant u \text{ and } V\leqslant v)}{\varepsilon}\end{aligned}$$

$$
\begin{aligned}
&= \lim_{\varepsilon\to 0} \frac{\mathbb{P}(u \leqslant U \leqslant u+\varepsilon \text{ and } V \leqslant v)}{P(u \leqslant U \leqslant u+\varepsilon)} \\
&= \lim_{\varepsilon\to 0} \mathbb{P}(V \leqslant v \mid u \leqslant U \leqslant u+\varepsilon) \\
&= \mathbb{P}(V \leqslant v \mid U = u) \\
&\leqslant 1,
\end{aligned}
$$

$u, v \in [0,1]$, hence

$$
h'(u) = \frac{\partial C}{\partial u}(u,v) - 1 = \mathbb{P}(V \leqslant v \mid U = u) - 1 \leqslant 0,
$$

$u, v \in [0,1]$, and since $h(1) = C(1,v) - v = \mathbb{P}(V \leqslant v) - v = 0$, $v \in [0,1]$ we conclude that $h(u) \geqslant 0$, $u \in [0,1]$, which shows (4.10).

Exercise 4.2

a) When $\rho = 1$, we have

$$
\begin{cases}
\mathbb{P}(X = 1 \text{ and } Y = 1) = p_X p_Y + \sqrt{p_X p_Y (1-p_X)(1-p_Y)}, \\
\mathbb{P}(X = 0 \text{ and } Y = 1) = (1-p_X)p_Y - \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \\
\mathbb{P}(X = 1 \text{ and } Y = 0) = p_X(1-p_Y) - \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \\
\mathbb{P}(X = 0 \text{ and } Y = 0) = (1-p_X)(1-p_Y) + \sqrt{p_X p_Y (1-p_X)(1-p_Y)},
\end{cases}
$$

hence

$$
\begin{cases}
(1-p_X)p_Y \geqslant \sqrt{p_X p_Y (1-p_X)(1-p_Y)}, \\
p_X(1-p_Y) \geqslant \sqrt{p_X p_Y (1-p_X)(1-p_Y)},
\end{cases}
$$

hence

$$
(1-p_X)p_Y \geqslant p_X(1-p_Y) \quad \text{and} \quad p_X(1-p_Y) \geqslant p_Y(1-p_X),
$$

showing that $(1-p_X)p_Y = p_X(1-p_Y)$, which implies $p_X = p_Y$, and

$$
\begin{cases}
\mathbb{P}(X = 1 \text{ and } Y = 1) = p_X^2 + p_X(1-p_X) = p_X = p_Y, \\
\mathbb{P}(X = 0 \text{ and } Y = 1) = 0, \\
\mathbb{P}(X = 1 \text{ and } Y = 0) = 0, \\
\mathbb{P}(X = 0 \text{ and } Y = 0) = 1 - p_X = 1 - p_Y.
\end{cases}
$$

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b) When $\rho = -1$, we have

$$\begin{cases} \mathbb{P}(X = 1 \text{ and } Y = 1) = p_X p_Y - \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \\ \mathbb{P}(X = 0 \text{ and } Y = 1) = (1-p_X) p_Y + \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \\ \mathbb{P}(X = 1 \text{ and } Y = 0) = p_X (1-p_Y) + \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \\ \mathbb{P}(X = 0 \text{ and } Y = 0) = (1-p_X)(1-p_Y) - \sqrt{p_X p_Y (1-p_X)(1-p_Y)} \geqslant 0, \end{cases}$$

hence

$$\begin{cases} p_X p_Y \geqslant \sqrt{p_X p_Y (1-p_X)(1-p_Y)}, \\ p_X p_Y \geqslant \sqrt{p_X p_Y (1-p_X)(1-p_Y)}, \end{cases}$$

hence

$$p_X p_Y \geqslant (1-p_X)(1-p_Y) \quad \text{and} \quad p_X p_Y \geqslant (1-p_X)(1-p_Y),$$

showing that $p_X p_Y = (1-p_X)(1-p_Y)$, which implies $p_X = 1 - p_Y$, and

$$\begin{cases} \mathbb{P}(X = 1 \text{ and } Y = 1) = 0, \\ \mathbb{P}(X = 0 \text{ and } Y = 1) = 1, \\ \mathbb{P}(X = 1 \text{ and } Y = 0) = 1, \\ \mathbb{P}(X = 0 \text{ and } Y = 0) = 0. \end{cases}$$

Exercise 4.3

a) We have

$$\mathbb{P}(X \geqslant x) = \mathbb{P}(X \geqslant x \text{ and } Y \geqslant 0) = \mathrm{e}^{-(\lambda+\nu)x},$$

and

$$\mathbb{P}(Y \geqslant y) = \mathbb{P}(X \geqslant 0 \text{ and } Y \geqslant y) = \mathrm{e}^{-(\mu+\nu)y},$$

$x, y \geqslant 0$, *i.e.* X and Y are exponentially distributed with respective parameters $\lambda + \nu$ and $\mu + \nu$.

b) We have

$$\begin{aligned} &\mathbb{P}(X \leqslant x \text{ and } Y \leqslant y) \\ &= 1 - \mathbb{P}(X \geqslant x) - \mathbb{P}(Y \geqslant y) + \mathbb{P}(X \geqslant x \text{ and } Y \geqslant y) \end{aligned}$$

$$= 1 - \mathrm{e}^{-(\lambda+\nu)x} - \mathrm{e}^{-(\mu+\nu)y} + \mathrm{e}^{-\lambda x-\mu y-\nu \operatorname{Max}(x,y)}, \quad x, y \geqslant 0.$$

c) We note that

$$\mathbb{P}\big(\mathrm{e}^{-(\lambda+\nu)X} \leqslant u\big) = \mathbb{P}\left(X \geqslant -\frac{\log u}{\lambda+\nu}\right) = u,$$

and similarly

$$\mathbb{P}\big(\mathrm{e}^{-(\mu+\nu)Y} \geqslant u\big) = \mathbb{P}\left(Y \geqslant -\frac{\log u}{\mu+\nu}\right) = u, \quad u \in [0,1],$$

hence the random variables $\mathrm{e}^{-(\lambda+\nu)X}$ and $\mathrm{e}^{-(\mu+\nu)Y}$ are uniformly distributed on $[0,1]$. Therefore, a copula function $C(u,v)$ can be defined by letting

$$\begin{aligned}
C(u,v) &:= \mathbb{P}\big(\mathrm{e}^{-(\lambda+\nu)X} \leqslant u \text{ and } \mathrm{e}^{-(\mu+\nu)Y} \leqslant v\big) \\
&= \mathbb{P}\left(X \geqslant -\frac{\log u}{\lambda+\nu} \text{ and } Y \geqslant -\frac{\log v}{\mu+\nu}\right) \\
&= \mathrm{e}^{\lambda(\lambda+\nu)^{-1}\log u+\mu(\mu+\nu)^{-1}\log v-\nu \operatorname{Max}(-(\lambda+\nu)^{-1}\log u,-(\mu+\nu)^{-1}\log v))} \\
&= u^{\lambda/(\lambda+\nu)} v^{\mu/(\mu+\nu)} \mathrm{e}^{-\nu \operatorname{Max}(-(\lambda+\nu)^{-1}\log u,-(\mu+\nu)^{-1}\log v))} \\
&= u^{\lambda/(\lambda+\nu)} v^{\mu/(\mu+\nu)} \mathrm{e}^{\nu \min(\log u^{1/(\lambda+\nu)},\log v^{1/(\mu+\nu)}))} \\
&= u^{\lambda/(\lambda+\nu)} v^{\mu/(\mu+\nu)} \mathrm{e}^{\log \min(u^{\nu/(\lambda+\nu)},v^{\nu/(\mu+\nu)}))} \\
&= u^{\lambda/(\lambda+\nu)} v^{\mu/(\mu+\nu)} \min(u^{\nu/(\lambda+\nu)}, v^{\nu/(\mu+\nu)})), \qquad x, y \geqslant 0.
\end{aligned}$$

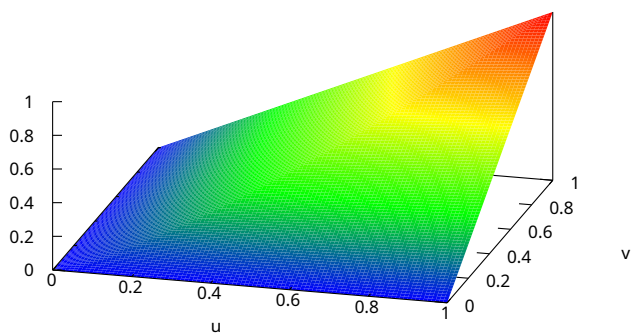


Fig. S.4: Exponential copula function $u, v \mapsto C(u,v)$ with $\lambda = 1$, $\mu = 2$, $\nu = 4$.

Exercise 4.4

a) We have

$$F_X(x) = \mathbb{P}(X \leqslant x) = \mathbb{P}(X \leqslant x \text{ and } Y \leqslant \infty) = \frac{1}{1+\mathrm{e}^{-x}}$$

and

$$F_Y(y) = \mathbb{P}(Y \leqslant y) = \mathbb{P}(X \leqslant \infty \text{ and } Y \leqslant y) = \frac{1}{1+\mathrm{e}^{-y}}, \qquad x, y \in \mathbb{R}.$$

The probability densities are given by

$$f_X(x) = f_Y(x) = F_X'(x) = F_Y'(x) = \frac{\mathrm{e}^{-x}}{(1+\mathrm{e}^{-x})^2}, \qquad x \in \mathbb{R}.$$

b) We have

$$F_X^{-1}(u) = F_Y^{-1}(u) = -\log\frac{1-u}{u}, \qquad u \in (0,1),$$

and the corresponding copula is given by

$$\begin{aligned} C(u,v) &= F_{(X,Y)}(F_X^{-1}(u), F_Y^{-1}(v)) \\ &= F_{(X,Y)}\left(-\log\frac{1-u}{u}, -\log\frac{1-v}{v}\right) \\ &= \frac{1}{1+(1-u)/u+(1-v)/v} \\ &= \frac{1}{1+(1-u)/u+(1-v)/v} \\ &= \frac{uv}{u+v-uv}, \qquad u, v \in [0,1], \end{aligned}$$

which is a particular case of the Ali–Mikhail–Haq copula.

Exercise 4.5

a) We show that (X,Y) have Gaussian marginals $\mathcal{N}(0,\sigma^2)$ and $\mathcal{N}(0,\eta^2)$, according to the following computation:

$$\begin{aligned} \int_{-\infty}^{\infty} \widetilde{f}(x,y)dy &= \frac{1}{\pi\sigma\eta}\int_{-\infty}^{\infty} \mathbb{1}_{\mathbb{R}^2_- \cup \mathbb{R}^2_+}(x,y)\mathrm{e}^{-x^2/(2\sigma^2)-y^2/(2\eta^2)}dy \\ &= \frac{1}{\pi\sigma\eta}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_-}(x)\int_{-\infty}^{0}\mathrm{e}^{-y^2/(2\eta^2)}dy \\ &\quad + \frac{1}{\pi\sigma\eta}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_+}(x)\int_{0}^{\infty}\mathrm{e}^{-y^2/(2\eta^2)}dy \end{aligned}$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-x^2/(2\sigma^2)} \mathbb{1}_{\mathbb{R}_-}(x) + \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-x^2/(2\sigma^2)} \mathbb{1}_{\mathbb{R}_+}(x)$$
$$= \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-x^2/(2\sigma^2)}, \qquad x \in \mathbb{R}.$$

b) The couple (X, Y) does *not* have a joint Gaussian distribution, and its joint probability density function does *not* coincide with $f_\Sigma(x, y)$.

c) When $\sigma = \eta = 1$, for $a \geqslant 0$ the random variable $X + Y$ has the probability density function

$$\begin{aligned}
&\frac{\partial}{\partial a}\mathbb{P}(X+Y \leqslant a) = \frac{\partial}{\partial a}\int_{-\infty}^{\infty}\int_{-\infty}^{a-x} \widetilde{f}(x,y)dydx \\
&= \frac{\partial}{\partial a}\int_{-\infty}^{0}\int_{-\infty}^{a-x} \widetilde{f}(x,y)dydx + \frac{\partial}{\partial a}\int_{0}^{\infty}\int_{-\infty}^{a-x} \widetilde{f}(x,y)dydx \\
&= \frac{\partial}{\partial a}\int_{-\infty}^{0}\int_{-\infty}^{0} \widetilde{f}(x,y)dydx + \frac{\partial}{\partial a}\int_{0}^{\infty}\int_{0}^{a-x} \widetilde{f}(x,y)dydx \\
&= \frac{1}{\pi}\frac{\partial}{\partial a}\int_0^a\int_0^{a-x} \mathrm{e}^{-x^2/2-y^2/2}dydx \\
&= \frac{1}{\pi}\frac{\partial}{\partial a}\int_0^a \mathrm{e}^{-(a-z)^2/2}\int_0^z \mathrm{e}^{-y^2/2}dydz \\
&= \frac{1}{\pi}\int_0^a \mathrm{e}^{-y^2/2}dy - \frac{1}{\pi}\int_0^a (a-z)\mathrm{e}^{-(a-z)^2/2}\int_0^z \mathrm{e}^{-y^2/2}dydz \\
&= \frac{1}{\pi}\int_0^a \mathrm{e}^{-y^2/2}dy - \frac{1}{\pi}\int_0^a (a-z)\mathrm{e}^{-(a-z)^2/2}dz\int_0^a \mathrm{e}^{-y^2/2}dy \\
&\quad + \frac{1}{\pi}\int_0^a \mathrm{e}^{-y^2/2}\int_0^y \mathrm{e}^{-(a-z)^2/2}dzdy \\
&= \frac{1}{\pi}\mathrm{e}^{-a^2/2}\int_0^a \mathrm{e}^{-y^2/2}dy + \frac{1}{\pi}\int_0^a \mathrm{e}^{-y^2/2}(\mathrm{e}^{-(a-y)^2/2} - \mathrm{e}^{-a^2/2})dy \\
&= \frac{1}{\pi}\mathrm{e}^{-a^2/2}\int_0^a \mathrm{e}^{-y^2/2-(a-y)^2/2}dy \\
&= \frac{1}{\pi}\int_0^a \mathrm{e}^{-((\sqrt{2}y-a/\sqrt{2})^2-a^2/2)/2}dy \\
&= \frac{\mathrm{e}^{-a^2/4}}{\pi\sqrt{2}}\int_0^{a\sqrt{2}} \mathrm{e}^{-((y-a/\sqrt{2})^2)/2}dy \\
&= \frac{\mathrm{e}^{-a^2/4}}{\pi\sqrt{2}}\int_{-a/\sqrt{2}}^{a(\sqrt{2}-1/\sqrt{2})} \mathrm{e}^{-y^2/2}dy \\
&= \frac{\mathrm{e}^{-a^2/4}}{\sqrt{\pi}\sqrt{2\pi}}\int_{-a/\sqrt{2}}^{a/\sqrt{2}} \mathrm{e}^{-y^2/2}dy \\
&= \mathrm{e}^{-a^2/4}\big(2\Phi\big(a/\sqrt{2}\big) - 1\big), \quad a \geqslant 0,
\end{aligned}$$

which vanishes at $a = 0$.

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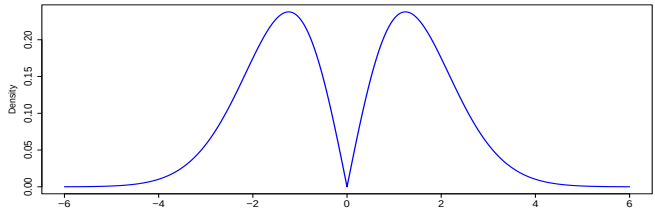


Fig. S.5: Density of $X+Y$.

d) The random variables X and Y are positively correlated, as

$$\begin{aligned}\int_{-\infty}^{\infty} y f_\Sigma(x,y)dy &= \frac{1}{\pi\sigma\eta}\int_{-\infty}^{\infty} \mathbb{1}_{\mathbb{R}_-^2\cup\mathbb{R}_+^2}(x,y) y \mathrm{e}^{-x^2/(2\sigma^2)-y^2/(2\eta^2)}dy \\ &= \frac{1}{\pi\sigma\eta}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_+}(x)\int_0^{\infty} y\mathrm{e}^{-y^2/(2\eta^2)}dy \\ &\quad + \frac{1}{\pi\sigma\eta}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_-}(x)\int_{-\infty}^{0} y\mathrm{e}^{-y^2/(2\eta^2)}dy \\ &= \frac{\eta}{\pi\sigma}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_+}(x) - \frac{\eta}{\pi\sigma}\mathrm{e}^{-x^2/(2\sigma^2)}\mathbb{1}_{\mathbb{R}_-}(x),\end{aligned}$$

hence

$$\begin{aligned}\mathbb{E}[XY] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} xy f_\Sigma(x,y)dydx \\ &= \frac{\eta}{\pi\sigma}\int_0^{\infty} x\mathrm{e}^{-x^2/(2\sigma^2)}dx - \frac{\eta}{\pi\sigma}\int_{-\infty}^{0} x\mathrm{e}^{-x^2/(2\sigma^2)}dx \\ &= \frac{2\sigma\eta}{\pi},\end{aligned}$$

and

$$\rho = \frac{\mathbb{E}[XY]}{\sigma\eta} = \frac{2}{\pi}.$$

Under a rotation

$$R = \begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix},$$

of angle $\theta\in[0,2\pi]$ we would find

$$\begin{aligned}&\mathbb{E}[(X\cos\theta - Y\sin\theta)(X\sin\theta + Y\cos\theta)] \\ &= \sin\theta\cos\theta\mathbb{E}\left[X^2\right] + (\cos^2\theta - \sin^2\theta)\mathbb{E}[XY] - \sin\theta\cos\theta\mathbb{E}\left[Y^2\right] \\ &= \sigma^2\sin\theta\cos\theta + (\cos^2\theta - \sin^2\theta)\frac{2\sigma\eta}{\pi} - \eta^2\sin\theta\cos\theta\end{aligned}$$

$$= \frac{\sigma^2}{2}\sin(2\theta) + \cos(2\theta)\frac{2\sigma\eta}{\pi} - \frac{\eta^2}{2}\sin(2\theta),$$

and

$$\rho = \frac{\sigma}{2\eta}\sin(2\theta) + \cos(2\theta)\frac{2}{\pi} - \frac{\eta}{2\sigma}\sin(2\theta),$$

i.e. $\theta = \pi/4$ and $\sigma = \eta$ would lead to uncorrelated random variables.

e) When $\sigma = \eta = 1$, the random variable $X^2 + Y^2$ has the cumulative distribution function

$$\begin{aligned}
\mathbb{P}(X^2+Y^2 \leqslant a) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathbf{1}_{\{x^2+y^2\leqslant a\}} \widetilde{f}(x,y)dydx \\
&= 2\int_0^{\infty}\int_0^{\infty} \mathbf{1}_{\{x^2+y^2\leqslant a\}} \widetilde{f}(x,y)dydx \\
&= \frac{2}{\pi}\int_0^{\infty}\int_0^{\infty} \mathbf{1}_{\{x^2+y^2\leqslant a\}} \mathrm{e}^{-x^2/2-y^2/2}dydx \\
&= \frac{2}{\pi}\int_0^{\pi/2}\int_0^{\sqrt{a}} \mathrm{e}^{-r^2/2} r dr d\theta \\
&= \int_0^{\sqrt{a}} \mathrm{e}^{-r^2/2} r dr \\
&= \Big[-\mathrm{e}^{-r^2/2}\Big]_0^{\sqrt{a}} \\
&= 1 - \mathrm{e}^{-a/2}, \quad a \geqslant 0,
\end{aligned}$$

where we used the change of variables in polar coordinates

$$r^2 = x^2 + y^2, \quad x = r\cos\theta, \quad y = r\sin\theta, \quad dxdy = rdrd\theta.$$

Hence, the random variable $X^2 + Y^2$ has an exponential distribution with parameter $1/2$.

Remark. This is also the distribution of $X^2 + Y^2$ when X and Y are independent.

Exercise 4.6

a) We have

$$\begin{aligned}
\mathbb{P}(\tau_i \wedge \tau \geqslant s) &= \mathbb{P}(\tau_i \geqslant s \text{ and } \tau \geqslant s) \\
&= \mathbb{P}(\tau_i \geqslant s)\mathbb{P}(\tau \geqslant s) \\
&= \mathrm{e}^{-\lambda_i s}\mathrm{e}^{-\lambda s} \\
&= \mathrm{e}^{-(\lambda_i+\lambda)s}, \qquad s \geqslant 0,
\end{aligned}$$

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hence $\tau_i \wedge \tau$ is an exponentially distributed random variable with parameter $\lambda_i + \lambda$, $i = 1, 2$.

b) Next, we have

$$\begin{aligned}
\mathbb{P}(\tau_1 \wedge \tau > s \text{ and } \tau_2 \wedge \tau > t) &= \mathbb{P}(\tau_1 > s \text{ and } \tau > s \text{ and } \tau_2 > t \text{ and } \tau > t) \\
&= \mathbb{P}(\tau_1 > s \text{ and } \tau_2 > t \text{ and } \tau > \operatorname{Max}(s,t)) \\
&= \mathbb{P}(\tau_1 > s)\mathbb{P}(\tau_2 > t)\mathbb{P}(\tau > \operatorname{Max}(s,t)) \\
&= \mathrm{e}^{-\lambda_1 s}\mathrm{e}^{-\lambda_2 t}\mathrm{e}^{-\lambda \operatorname{Max}(s,t)} \\
&= \mathrm{e}^{-\lambda_1 s - \lambda_2 t - \lambda \operatorname{Max}(s,t)} \\
&= \mathrm{e}^{-(\lambda_1+\lambda)s - (\lambda_2+\lambda)t + \lambda \min(s,t)} \\
&= (1 - F_X(s))(1 - F_Y(t)) \min(\mathrm{e}^{\lambda s}, \mathrm{e}^{\lambda t}),
\end{aligned}$$

$s, t \geqslant 0$.

c) We have

$$\begin{aligned}
F_{X,Y}(s,t) &= \mathbb{P}(\tau_1 \wedge \tau \leqslant s \text{ and } \tau_2 \wedge \tau \leqslant t) \\
&= \mathbb{P}(\tau_1 \wedge \tau \leqslant s) - \mathbb{P}(\tau_1 \wedge \tau \leqslant s \text{ and } \tau_2 \wedge \tau > t) \\
&= \mathbb{P}(\tau_1 \wedge \tau \leqslant s) - \big(\mathbb{P}(\tau_2 \wedge \tau > t) - \mathbb{P}(\tau_1 \wedge \tau > s \text{ and } \tau_2 \wedge \tau > t)\big) \\
&= \mathbb{P}(\tau_1 \wedge \tau \leqslant s) + \mathbb{P}(\tau_2 \wedge \tau \leqslant t) + \mathbb{P}(\tau_1 \wedge \tau > s \text{ and } \tau_2 \wedge \tau > t) - 1 \\
&= F_X(s) + F_Y(t) + (1 - F_X(s))(1 - F_Y(t)) \min(\mathrm{e}^{\lambda s}, \mathrm{e}^{\lambda t}) - 1.
\end{aligned}$$

d) We find

$$\begin{aligned}
C(u,v) &= F_{X,Y}(F_X^{-1}(u), F_Y^{-1}(v)) \\
&= F_X(F_X^{-1}(u)) + F_Y(F_X^{-1}(v)) \\
&\quad + (1 - F_X(F_X^{-1}(u)))(1 - F_Y(F_X^{-1}(v))) \min\left(\mathrm{e}^{\lambda F_X^{-1}(u)}, \mathrm{e}^{\lambda F_X^{-1}(v)}\right) - 1 \\
&= u + v - 1 + (1-u)(1-v) \min\left(\mathrm{e}^{\lambda F_X^{-1}(u)}, \mathrm{e}^{\lambda F_Y^{-1}(v)}\right) \\
&= u + v - 1 + (1-u)(1-v) \min\left(\mathrm{e}^{-\lambda \log(1-u)/(\lambda_1+\lambda)}, \mathrm{e}^{-\lambda \log(1-v)/(\lambda_2+\lambda)}\right) \\
&= u + v - 1 + \min\left((1-v)(1-u)^{1-\lambda/(\lambda_1+\lambda)}, (1-u)(1-v)^{1-\lambda/(\lambda_2+\lambda)}\right) \\
&= u + v - 1 + \min\left((1-v)(1-u)^{1-\theta_1}, (1-u)(1-v)^{1-\theta_2}\right), \quad u, v \in [0,1],
\end{aligned}$$

with

$$\theta_1 = \frac{\lambda}{\lambda_1 + \lambda} \quad \text{and} \quad \theta_2 = \frac{\lambda}{\lambda_2 + \lambda}.$$

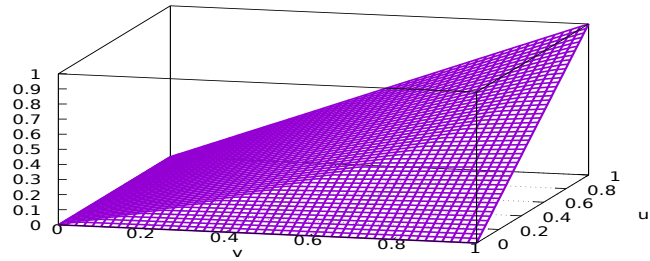


Fig. S.6: Survival copula graph with $\theta_1 = 0.3$ and $\theta_2 = 0.7$.

e) We have

$$C(u,v) = u + v - 1 + (1-u)(1-v)^{1-\theta_2}\mathbb{1}_{\{(1-u)^{\theta_1}<(1-v)^{\theta_2}\}} + (1-v)(1-u)^{1-\theta_1}\mathbb{1}_{\{(1-u)^{\theta_1}>(1-v)^{\theta_2}\}}, \quad u,v \in [0,1],$$

hence

$$\frac{\partial C}{\partial u}(u,v) = -(1-v)^{1-\theta_2}\mathbb{1}_{\{(1-u)^{\theta_1}<(1-v)^{\theta_2}\}} - (1-\theta_1)(1-v)(1-u)^{-\theta_1}\mathbb{1}_{\{(1-u)^{\theta_1}>(1-v)^{\theta_2}\}}$$

and the survival copula density is given by

$$\frac{\partial^2 C}{\partial u \partial v}(u,v) = (1-\theta_2)(1-v)^{-\theta_2}\mathbb{1}_{\{(1-u)^{\theta_1}<(1-v)^{\theta_2}\}} + (1-\theta_1)(1-u)^{-\theta_1}\mathbb{1}_{\{(1-u)^{\theta_1}>(1-v)^{\theta_2}\}}, \quad u,v \in [0,1],$$

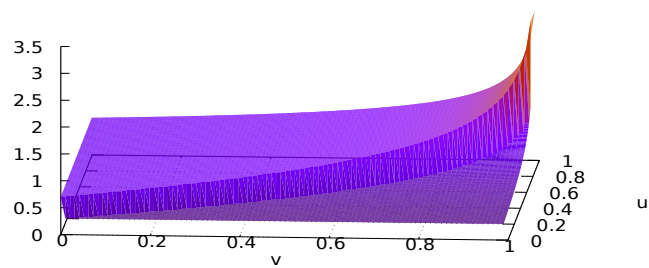


Fig. S.7: Survival copula density graph with $\theta_1 = 0.3$ and $\theta_2 = 0.7$.

Remark: When $\lambda = 0$ we have $\theta_1 = \theta_2 = 0$ and $\tau = +\infty$ *a.s.*, therefore we have

$$\min(\tau_1, \tau) = \tau_1 \quad \text{and} \quad \min(\tau_2, \tau) = \tau_2,$$

hence the copula $C(u, v)$ is given by

$$C(u, v) = u + v - 1 + (1 - v)(1 - u) = uv, \qquad u, v \in [0, 1],$$

which coincides with the copula of independence.

Chapter 5

Exercise 5.1 The payoff C is that of a *put* option with strike price $K = \$3$.

Exercise 5.2 Each of the two possible scenarios yields one equation:

$$\begin{cases} 5\xi + \eta = 0 \\ 2\xi + \eta = 6, \end{cases} \quad \text{with solution} \quad \begin{cases} \xi = -2 \\ \eta = +10. \end{cases}$$

The hedging strategy at $t = 0$ is to **shortsell** $-\xi = +2$ units of the asset S priced $S_0 = 4$, and to put $\eta = \$10$ on the savings account. The price $V_0 = \xi S_0 + \eta$ of the initial portfolio at time $t = 0$ is

$$V_0 = \xi S_0 + \eta = -2 \times 4 + 10 = \$2,$$

which yields the price of the claim at time $t = 0$. In order to hedge then option, one should:

i) At time $t = 0$,

   a. Charge the \$2 option price.
   b. Shortsell $-\xi = +2$ units of the stock priced $S_0 = 4$, which yields \$8.
   c. Put $\eta = \$8 + \$2 = \$10$ on the savings account.

ii) At time $t = 1$,

   a. If $S_1 = \$5$, spend \$10 from savings to buy back $-\xi = +2$ stocks.
   b. If $S_1 = \$2$, spend \$4 from savings to buy back $-\xi = +2$ stocks, and deliver a \$10 - \$4 = \$6 payoff.

Pricing the option by the expected value $\mathbb{E}^*[C]$ yields the equality

$$\$2 = \mathbb{E}^*[C]$$

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$$
\begin{aligned}
&= 0 \times \mathbb{P}^*(C = 0) + 6 \times \mathbb{P}^*(C = 6) \\
&= 0 \times \mathbb{P}^*(S_1 = 2) + 6 \times \mathbb{P}^*(S_1 = 5) \\
&= 6 \times q^*,
\end{aligned}
$$

hence the risk-neutral probability measure $\mathbb{P}^*$ is given by

$$
p^* = \mathbb{P}^*(S_1 = 5) = \frac{2}{3} \quad \text{and} \quad q^* = \mathbb{P}^*(S_1 = 2) = \frac{1}{3}.
$$

Exercise 5.3

a) Each of the stated conditions yields one equation, *i.e.*

$$
\begin{cases} 4\xi + \eta = 1 \\ 5\xi + \eta = 3, \end{cases} \qquad \text{with solution} \qquad \begin{cases} \xi = 2 \\ \eta = -7. \end{cases}
$$

Therefore, the portfolio allocation at $t = 0$ consists to purchase $\xi = 2$ unit of the asset S priced $S_0 = 4$, and to borrow $-\eta = \$7$ in cash.

We can check that the price $V_0 = \xi S_0 + \eta$ of the initial portfolio at time $t = 0$ is

$$
V_0 = \xi S_0 + \eta = 2 \times 4 - 7 = \$1.
$$

b) This loss is expressed as

$$
\xi \times \$2 + \eta = 2 \times 2 - 7 = -\$3.
$$

Note that the \$1 received when selling the option is not counted here because it has already been fully invested into the portfolio.

Exercise 5.4

a) i) Does this model allow for arbitrage? Yes | ✓ | No |

ii) If this model allows for arbitrage opportunities, how can they be realized? By shortselling | | By borrowing on savings | ✓ | N.A. |

b) i) Does this model allow for arbitrage? Yes | | No | ✓

ii) If this model allows for arbitrage opportunities, how can they be realized? By shortselling | | By borrowing on savings | | N.A. | ✓

c) i) Does this model allow for arbitrage? Yes | ✓ | No |

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ii) If this model allows for arbitrage opportunities, how can they be realized? | By shortselling | ✓ | By borrowing on savings | | N.A. |

Exercise 5.5 Hedging a claim with possible payoff values C_a, C_b, C_c would require to solve

$$\begin{cases} (1+a)\xi S_0^{(1)} + (1+r)\eta S_0^{(0)} = C_a \\ (1+b)\xi S_0^{(1)} + (1+r)\eta S_0^{(0)} = C_b \\ (1+c)\xi S_0^{(1)} + (1+r)\eta S_0^{(0)} = C_c, \end{cases}$$

for ξ and η, which is not possible in general due to the existence of three conditions with only two unknowns.

Exercise 5.6 We check that for any $\mathbb{P}^*$ of the form

$$\mathbb{P}^*(R_t = -1) := p^*, \quad \mathbb{P}^*(R_t = 0) := 1 - 2p^*, \quad \mathbb{P}^*(R_t = 1) := p^*,$$

we have

$$\mathbb{E}^*[S_1] = S_0(2p^* + 1 - 2p^*) = S_0,$$

and similarly

$$\mathbb{E}^*[S_2 \mid S_1] = S_1(2p^* + (1 - 2p^*)) = S_1,$$

hence the probability measure $\mathbb{P}^*$ is risk-neutral.

Exercise 5.7

a) Every eligible Tencent shareholder received the equivalent of

$$\frac{\text{HK\$281.40}}{21} = \text{HK\$13.4}.$$

b) For the *put warrant holder*, the anticipated payoff is $(K - 1.03 \times S_T)^+/100$ with K=HK\$426.06, see HSB Ltd. (2021), *i.e.*

$$\begin{aligned} \frac{1}{100}(K - 1.03 \times S_T)^+ &= \frac{1.03}{100}(0.97087 \times K - S_T)^+ \\ &= \frac{1}{97.087}(0.97087 \times K - S_T)^+. \end{aligned}$$

Therefore, after adjusting the put option strike price to

$$0.97087 \times K = 0.97087 \times \text{HK\$426.06} = \text{HK\$413.65},$$

the warrant holder will own $1/97.087 = 0.0103$ options. In other words, one option is now worth 97.087 warrants, which is the adjusted entitlement ratio.

c) For the *call warrant holder*, the anticipated payoff is $(1.03 \times S_T - K)^+/100$, *i.e.*

$$\begin{aligned}\frac{1}{100}(1.03 \times S_T - K)^+ &= \frac{1.03}{100}(S_T - 0.97087 \times K)^+ \\ &= \frac{1}{97.087}(S_T - 0.97087 \times K)^+,\end{aligned}$$

hence the adjustment in strike price and entitlement ratio is the same as for put options.

Exercise 5.8

a) Each of two possible scenarios yields one equation:

$$\begin{cases} \alpha \overline{S}_1 + \beta = \overline{S}_1 - K \\ \alpha \underline{S}_1 + \beta = 0, \end{cases} \quad \text{with solution} \quad \begin{cases} \alpha = \dfrac{\overline{S}_1 - K}{\overline{S}_1 - \underline{S}_1} \\ \beta = -\underline{S}_1 \dfrac{\overline{S}_1 - K}{\overline{S}_1 - \underline{S}_1}. \end{cases}$$

b) We have

$$0 \leqslant \alpha = \frac{\overline{S}_1 - K}{\overline{S}_1 - \underline{S}_1} \leqslant 1$$

since $K \in [\underline{S}_1, \overline{S}_1]$.

c) We find

$$\begin{aligned}\mathrm{SRM_C} &= \alpha S_0 + \beta \\ &= \alpha(S_0 - \underline{S}_1) \\ &= (S_0 - \underline{S}_1)\frac{\overline{S}_1 - K}{\overline{S}_1 - \underline{S}_1}.\end{aligned}$$

We note that when $S_0 < \underline{S}_1$ the value of $\mathrm{SRM_C}$ is negative because in this case, investing in the zero-cost portfolio $(\alpha, -\alpha S_0)$ that would yield a payoff at least equal to $\alpha(\underline{S}_1 - S_0) = -\alpha(\underline{S}_0 - S_1) > 0$, which represents an *arbitrage opportunity*.

Exercise 5.9

a) This range forward contract can be realized by

- holding one unit of S_T,

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- holding one put option with strike price K_1,
- shorting one call option with strike price K_2,
- borrowing $\$F$ in cash.

b) The graph of the payoff function of the range forward contract with $K_1 = \$80$, $F = \$100$, $K_2 = \$110$ is as follows:

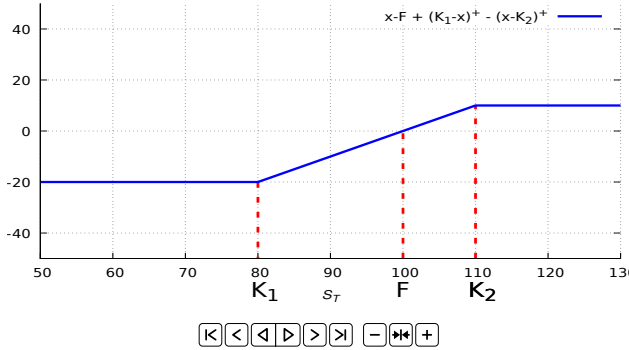


Fig. S.8: Range forward contract payoff as a combination of call and put payoffs.* See also http://optioncreator.com/st7ulpk.

Exercise 5.10 The risk-neutral measure is given by

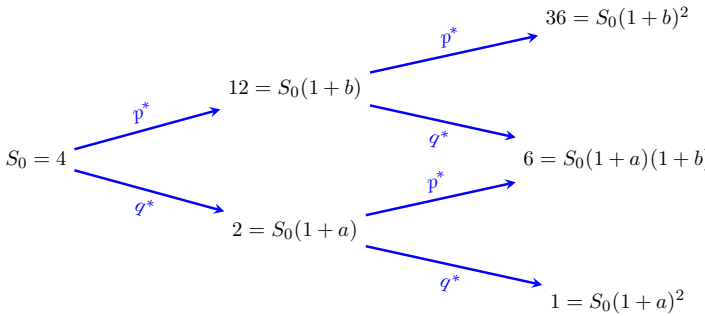
$$p^* = \frac{r-a}{b-a} = \frac{1-0.5}{2-(-0.5)} = \frac{3}{5}$$

and

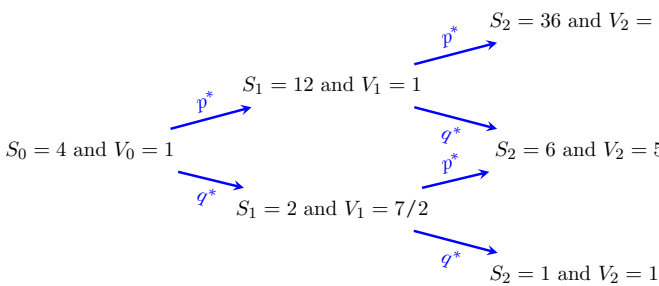
$$q^* = \frac{b-r}{b-a} = \frac{2-1}{2-(-0.5)} = \frac{2}{5},$$

and we have the binary tree

* The animation works in Acrobat Reader on the entire pdf file.



and



and the table

$S_0 = 4$ $V_0 = 1$	$S_1 = 12$, $V_1 = 1$	$S_2 = 36$ $V_2 = 0$
		$S_2 = 6$ $V_2 = 5$
	$S_1 = 2$, $V_1 = 7/2$	$S_2 = 1$ $V_2 = 11$

Table S.2: CRR pricing tree.

This can be summarized in the following table:

	$S_1 = 12, V_1 = 1$	$S_2 = 36$
$S_0 = 4$		$V_2 = 0$
$V_0 = 1$	$\xi_2 = -1/6, \eta_2 = 3/2$	$S_2 = 6$
$\xi_1 = -1/4$	$S_1 = 2, V_1 = 7/2$	$V_2 = 5$
$\eta_1 = 2$		$S_2 = 1$
	$\xi_2 = -1, \eta_2 = 11/4$	$V_2 = 10$

Table S.3: CRR pricing and hedging tree.

We also check that the self-financing condition

$$\xi_1 S_1 + \eta_1 A_1 = \xi_2 S_1 + \eta_2 A_1$$

is satisfied. The attached IPython **notebook*** can be used to recover the above results.

Exercise 5.11

a) The payoff of the long box spread option is given in terms of K_1 and K_2 as

$$\begin{aligned}(x-K_1)^+ - (K_1-x)^+ - (x-K_2)^+ + (K_2-x)^+ &= x - K_1 - (x-K_2)\\ &= K_2 - K_1.\end{aligned}$$

b) From Table 5.1 we check that the strike prices suitable for a long box spread option on the Hang Seng Index (HSI) are $K_1 = 25,000$ and $K_2 = 25,200$.

c) Based on the data provided, we note that the long box spread can be realized in two ways.

i) Using the put option issued by BI (BOCI Asia Ltd.) at 0.044.

In this case, the box spread option represents a short position priced

$$\underbrace{0.540}_{\text{Long call}} \times 7,500 \; \underbrace{-0.064}_{\text{Short put}} \times 8,000 \; \underbrace{-0.370}_{\text{Short call}} \times 11,000 \; \underbrace{+0.044}_{\text{Long put}} \times 10,000 = -92$$

index points, or $-92 \times \$50 = -\$4,600$.

Note that option prices are quoted in index points (to be multiplied by the relevant option/warrant entitlement ratio), and every index point is worth \$50.

ii) Using the put option issued by HT (Haitong Securities) at 0.061.

In this case, the box spread option represents a long position priced

* Right-click to save as attachment (may not work on 🍎).

$$\underbrace{0.540}_{\text{Long call}} \times 7,500 \underbrace{-0.044}_{\text{Short put}} \times 8,000 \underbrace{-0.370}_{\text{Short call}} \times 11,000 \underbrace{+0.061}_{\text{Long put}} \times 10,000 = +78$$

index points, or $78 \times \$50 = \$3,900$.

d) As the option built in ci) represents a short position paying $\$4,600$ today with an additional $\$50 \times (K_2 - K_1) = 200 = \$10,000$ payoff at maturity on March 28, I would definitely enter this position.

As for the option built in cii), it is less profitable because it costs $\$3,900$, however it is still profitable taking into account the $\$10,000$ payoff at maturity on March 28.

Exercise 5.12

a) At times $t = 0, 1$, the value of the quantity X_t of asset X quoted in units of Y is

$$X_t S_t = Y_t,$$

and we have

$$\mathrm{LP}_t = Y_t + X_t S_t = 2Y_t = 2\sqrt{CS_t}.$$

We note that in the CPAMM model, the dollar amounts in X and Y remaining in the pool at times $t = 0, 1$ are both equal to $Y_t = CS_t$.

b) We have

$$V_t = Y_0 + X_0 S_t = \sqrt{CS_0}\left(1 + \frac{S_t}{S_0}\right), \quad t = 0, 1.$$

c) We have

$$V_t - \mathrm{LP}_t = \sqrt{CS_0}\left(1 + \frac{S_t}{S_0}\right) - 2\sqrt{CS_t} = \sqrt{CS_0}\left(1 - \sqrt{\frac{S_t}{S_0}}\right)^2 \geqslant 0, \quad t = 0, 1.$$

d) We need to satisfy the conditions

$$\begin{cases} V_1 - \mathrm{LP}_1 = \xi|a - S_0| & \text{if } S_1 = a, \\ V_1 - \mathrm{LP}_1 = \xi|b - S_0| & \text{if } S_1 = b, \end{cases}$$

i.e.

$$\begin{cases} \sqrt{CS_0}\left(1 - \sqrt{\dfrac{a}{S_0}}\right)^2 = \xi\left(1 - \dfrac{a}{S_0}\right) & \text{if } S_1 = a, \\ \sqrt{CS_0}\left(1 - \sqrt{\dfrac{b}{S_0}}\right)^2 = \xi\left(\dfrac{b}{S_0} - 1\right) & \text{if } S_1 = b, \end{cases}$$

hence

$$\xi = \sqrt{CS_0}\frac{(1-\sqrt{a/S_0})^2}{1-a/S_0} = \sqrt{CS_0}\frac{(\sqrt{b/S_0}-1)^2}{b/S_0-1}.$$

Exercise 5.13

a) 1) We have $b_{k,l} = d_k p^*_{k,l}$, $k,l = 1,2$.

2) We have

$$d_k = d_k(p^*_{k,1} + p^*_{k,2}) = b_{k,1} + b_{k,2}, \qquad k = 1,2.$$

3) We have

$$p^*_{k,l} = \frac{b_{k,l}}{d_k} = \frac{b_{k,l}}{b_{k,1}+b_{k,2}}, \qquad k,l = 1,2.$$

b) 1) There are 7 unknowns $\delta, u_1, u_2, p^*_{1,1}, p^*_{1,2}, p^*_{2,1}, p^*_{2,2}$ and $4+2=6$ equations.

2) We have

$$b_{k,1}u_1 + b_{k,2}u_2 = \delta u_k p_{k,1} + \delta u_k p_{k,2} = \delta u_k, \qquad k = 1,2.$$

3) Diagonalization shows that, up to a multiplicative constant, we have either:

i)

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} -\dfrac{\sqrt{(b_{1,1}-b_{1,2})^2 + 4b_{1,2}b_{2,1}} - (b_{1,1}-b_{2,2})}{2b_{2,1}} < 0 \\ 1 \end{bmatrix}$$

and

$$\delta = \frac{b_{1,1}+b_{2,2} - \sqrt{(b_{1,1}-b_{2,2})^2 + 4b_{1,2}b_{2,1}}}{2},$$

ii) or

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} \dfrac{b_{1,1}-b_{2,2} + \sqrt{(b_{1,1}-b_{2,2})^2 + 4b_{1,2}b_{2,1}}}{2b_{2,1}} > 0 \\ 1 \end{bmatrix}$$

and

$$\delta = \frac{b_{1,1}+b_{2,2} + \sqrt{(b_{1,1}-b_{2,2})^2 + 4b_{1,2}b_{2,1}}}{2} > 0,$$

and we check that only the second choice (ii) yields positive values.

4) We find

$$p_{1,1} = \frac{1}{\delta} b_{1,1} = \frac{2b_{1,1}}{b_{1,1} + b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}},$$

$$p_{1,2} = \frac{1}{\delta} b_{1,2} \frac{u_2}{u_1}$$

$$= \frac{4b_{1,2}b_{2,1}}{\left(b_{1,1} - b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}\right)\left(b_{1,1} + b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}\right)},$$

$$p_{2,1} = \frac{1}{\delta} b_{2,1} \frac{u_1}{u_2} = \frac{b_{1,1} - b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}}{b_{1,1} + b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}},$$

$$p_{2,2} = \frac{1}{\delta} b_{2,2} = \frac{2b_{2,2}}{b_{1,1} + b_{2,2} + \sqrt{(b_{1,1} - b_{2,2})^2 + 4b_{1,2}b_{2,1}}}.$$

Exercise 5.14

a) 1) We have $b_{k,l} = d_k p^*_{k,l}$, $k, l = 1, \ldots, N$.

2) We have

$$d_k = d_k \sum_{l=1}^{N} p^*_{k,l} = \sum_{l=1}^{N} b_{k,l}, \qquad k = 1, \ldots, N.$$

3) We have

$$p^*_{k,l} = \frac{b_{k,l}}{d_k} = \frac{b_{k,l}}{b_{k,1} + \cdots + b_{k,N}}, \qquad k, l = 1, \ldots, N,$$

which yields the equation $Bu^\top = \delta u^\top$.

b) 1) There are $N^2 + N + 1$ unknowns $\delta, u_1, \ldots, u_N, p^*_{k,l}$, $k, l = 1, \ldots, N$, and $N^2 + N$ equations.

2) We have

$$\sum_{l=1}^{N} b_{k,l} u_l = \delta u_k \sum_{l=1}^{N} p_{k,l} = \delta u_k, \qquad k = 1, \ldots, N.$$

3) We note that B is a positive matrix, in the sense that $b_{k,l} > 0$, $k, l = 1, \ldots, N$. Consequently, by the Perron–Frobenius theorem there exists a choice of eigenvalue δ and eigenvector $(u_1, \ldots, u_N)$ such that $\delta > 0$ and $u_k > 0$, $k = 1, \ldots, N$. Moreover, this choice is unique.

4) Once the unique set of parameters $\delta > 0$ and $u_k > 0$, $k = 1, \ldots, N$ has been determined, we use the relation

$$p_{k,l} = \frac{u_l b_{k,l}}{\delta u_k}$$

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to recover the transition probabilities $p_{k,l}$ from the binary option prices $b_{k,l}$, $k, l = 1, \ldots, N$.

5) In this case, the probabilities $p_{k,l} = p_k$ and $p^*_{k,l} = p^*_k$ do not depend on $l \in \{1, \ldots, N\}$ and we have $b_{k,l} = d_k p^*_{k,l} = d_k p^*_l$ and $u_l b_{k,l} = \delta u_k p_{k,l} = \delta u_k p_l$ hence the relation

$$u_l d_k p^*_l = \delta u_k p_l, \qquad k, l = 1, \ldots, N.$$

By summation over l, this gives $d_k \sum_{l=1}^N u_l p^*_l = \delta u_k$, hence

$$d_k = \frac{\delta u_k}{u_1 p^*_1 + \cdots + u_N p^*_N}, \qquad k = 1, \ldots, N.$$

If in addition $d_k = d$ does not depend on $k = 1, \ldots, N$, we find that

$$d = \frac{\delta u_k}{u_1 p^*_1 + \cdots + u_N p^*_N},$$

hence $u = u_k$ does not depend on $k = 1, \ldots, N$, and

$$d = \frac{\delta u_k}{u_1 p^*_1 + \cdots + u_N p^*_N} = \frac{\delta u}{u p^*_1 + \cdots + u p^*_N} = \delta,$$

and therefore $p^*_l = p_l$, $l = 1, \ldots, N$.

Chapter 6

Exercise 6.1

a) The expected value premium principle (6.2) is

- monotone since $X \leqslant Y$ implies $(1+\alpha)\mathbb{E}[X] \leqslant (1+\alpha)\mathbb{E}[Y]$,
- subadditive since $(1+\alpha)\mathbb{E}[X+Y] = (1+\alpha)\mathbb{E}[X] + (1+\alpha)\mathbb{E}[Y]$,
- positively homogeneous since $(1+\alpha)\mathbb{E}[\lambda X] = (1+\alpha)\lambda\mathbb{E}[X]$ for all $\lambda > 0$.

However, it is not translation invariant, since

$$\begin{aligned}(1+\alpha)\mathbb{E}[\mu + X] &= (1+\alpha)\mu + (1+\alpha)\mathbb{E}[X] \\ &\neq \mu + (1+\alpha)\mathbb{E}[X], \quad \mu \in \mathbb{R},\end{aligned}$$

unless $\alpha = 0$. Therefore, the expected value premium principle is not a coherent risk measure when $\alpha > 0$.

b) The standard deviation premium principle (6.3) is subadditive due to Lemma 6.18, and it satisfies the positive homogeneity and translation invariance properties. However, when $\alpha > 0$ it is not a coherent risk measure because it is not monotone. Indeed, if $Y = 0$ and X takes the values $(-x, 0)$ with probabilities $(p, 1-p)$ for some $x > 0$ and $p \in (0,1)$, then

$$\mathbb{E}[X] + \alpha\sqrt{\mathrm{Var}(X)} = -px + \alpha x\sqrt{(1-p)p},$$

hence

$$\mathbb{E}[X] + \alpha\sqrt{\mathrm{Var}(X)} > 0 = \mathbb{E}[Y] + \alpha\sqrt{\mathrm{Var}(Y)},$$

provided that

$$0 < p < \frac{\alpha^2}{1+\alpha^2}.$$

Exercise 6.2

a) We have

$$\begin{aligned} F_X(x) &= \mathbb{P}(X \leqslant x) \\ &= \int_0^x f_X(y)dy \\ &= \gamma\theta^\gamma \int_0^x \frac{1}{(\theta+y)^{\gamma+1}}dy \\ &= \left[-\left(\frac{\theta}{\theta+y}\right)^\gamma\right]_0^x \\ &= 1 - \left(\frac{\theta}{\theta+x}\right)^\gamma, \qquad x \in \mathbb{R}_+. \end{aligned}$$

b) Since the distribution of X admits a probability density function, the cumulative distribution function $x \longmapsto F_X(x)$ is continuous in x and we have $\mathbb{P}(X = x) = 0$ for all $x > 0$. Hence the Value at Risk V_X^p at the level p is given by the relation $F_X(V_X^p) = p$, *i.e.*

$$\left(\frac{\theta}{\theta + V_X^p}\right)^\gamma = 1 - p,$$

which gives

$$V_X^p = \theta\left(\frac{1}{(1-p)^{1/\gamma}} - 1\right).$$

In particular, with $p = 99\%$, $\theta = 40$ and $\gamma = 2$, we find

$$V_X^p = ((1-p)^{-1/\gamma} - 1)\theta = 40\big(\sqrt{100} - 1\big) = \$360.$$

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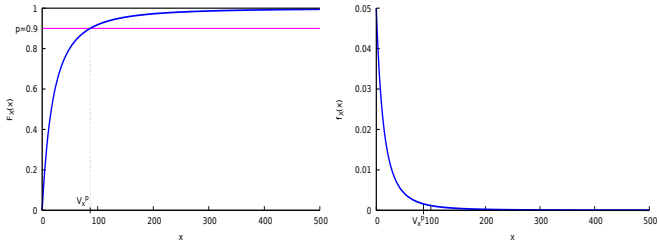


Fig. S.9: Pareto CDF $x \mapsto F_X(x)$ and PDF $x \mapsto f_X(x)$ with $V_X^{99\%} = \$86.49$.

Exercise 6.3

a) We have $\mathbb{P}(X = 100) = 0.02$.
b) We have $V_X^q = 100$ for all $q \in [0.97, 0.99]$.
c) The Value at Risk V_X^q at the level $q \in [0.99, 1]$ satisfies

$$F_X(V_X^q) = \mathbb{P}(X \leqslant V_X^q) = 0.99 + 0.01 \times (V_X^q - 100)/50 = q,$$

hence

$$V_X^q = 100 + 50(100q - 99) = 5000q - 4850, \qquad q \in [0.99, 1].$$

Exercise 6.4 We find $V_X^{99\%} = 100$ according to the following cumulative distribution function:

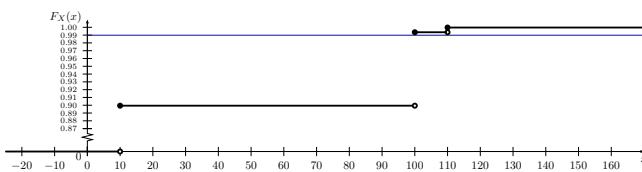


Fig. S.10: Cumulative distribution function of X and Y.

Exercise 6.5

a) We have

$$V_X^p := \inf\{x \in \mathbb{R} \ : \ \mathbb{P}(X \leqslant x) \geqslant p\} = -\frac{1}{\lambda}\log(1-p) = \mathbb{E}[X]\log\frac{1}{1-p}.$$

When $p = 95\%$ this yields

$$V_X^p \simeq 2.996\mathbb{E}[X].$$

b) We find that the required capital C_X satisfies

$$C_X = V_X^p - \mathbb{E}[X] = \mathbb{E}[X]\log\frac{1}{1-p} - \mathbb{E}[X],$$

i.e.

$$C_X = V_X^{95\%} - \mathbb{E}[X] \simeq 1.996\mathbb{E}[X],$$

which means doubling the estimated amount of liabilities.

Exercise 6.6 By Proposition 6.2 and the geometric series identity (13.54), we have

$$\begin{aligned}
\mathbb{E}[X \mid X \geqslant a] &= \frac{1}{\mathbb{P}(X \geqslant a)}\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant a\}}\big] \\
&= \frac{1}{\mathbb{P}(X \geqslant a)}\sum_{k\geqslant a} k\mathbb{P}(X=k) \\
&= \frac{1}{\sum_{k\geqslant a}(1-p)^k}\sum_{k\geqslant a} k(1-p)^k \\
&= \frac{(1-p)^a}{(1-p)^a\sum_{k\geqslant 0}(1-p)^k}\sum_{k\geqslant 0}(k+a)(1-p)^k \\
&= a + \frac{1}{\sum_{k\geqslant 0}(1-p)^k}\sum_{k\geqslant 0} k(1-p)^k \\
&= a + p\sum_{k\geqslant 0} k(1-p)^k \\
&= a + \frac{1}{p} \\
&= a + \mathbb{E}[X].
\end{aligned}$$

This can be recovered numerically for example with $a = 11$ using the following code.

```
geo_samples <- rgeom(100000, prob = 1/4)
mean(geo_samples)
mean(geo_samples[geo_samples>=10])
```

Listing S.11: code - Conditional geometric distribution mean.

```python
import numpy as np; geo_samples = np.random.geometric(p=1/4, size=100000)
print(f"Mean of all samples: {np.mean(geo_samples)}")
print(f"Mean of samples >= 10: {np.mean(geo_samples[geo_samples >= 10])}")
```

Listing S.12: Python code - Conditional geometric distribution mean.

Exercise 6.7

a) As in the proof of the Markov inequality, for every $x > 0$ and $r > 0$ we have

$$\begin{aligned} x^r \mathbb{P}(X \geqslant x) &= x^r \mathbb{E}\left[\mathbb{1}_{\{X \geqslant x\}}\right] \\ &\leqslant \mathbb{E}\left[X^r \mathbb{1}_{\{X \geqslant x\}}\right] \\ &\leqslant \mathbb{E}\left[|X|^r\right], \end{aligned}$$

hence

$$\mathbb{P}(X \leqslant x) \geqslant 1 - \frac{1}{x^r}\mathbb{E}[|X|^r], \qquad x > 0. \tag{A.1}$$

From the inequality (A.1), it follows that

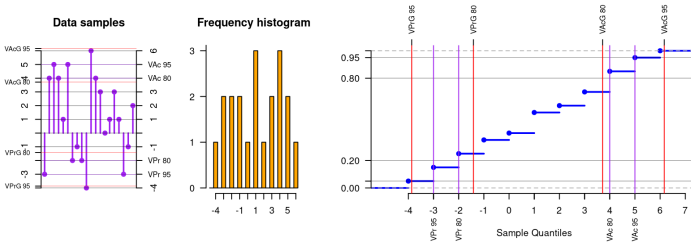
$$\begin{aligned} V_X^p &= \inf\left\{x \in \mathbb{R} \ : \ \mathbb{P}(X \leqslant x) \geqslant p\right\} \\ &\leqslant \inf\left\{x \in \mathbb{R} \ : \ 1 - \frac{1}{x^r}\mathbb{E}[|X|^r] \geqslant p\right\} \\ &= \inf\left\{x \in \mathbb{R} \ : \ x^r \geqslant \frac{1}{1-p}\mathbb{E}[|X|^r]\right\} \\ &= \left(\frac{\mathbb{E}[|X|^r]}{1-p}\right)^{1/r} \\ &= \frac{\|X\|_{L^r(\Omega)}}{(1-p)^{1/r}}. \end{aligned}$$

b) Taking $p = 95\%$ and $r = 1$, we get

$$V_X^{95\%} \leqslant \frac{1}{1-p}\mathbb{E}[|X|] = 20\mathbb{E}[|X|].$$

To summarize, a smaller L^r-norm of X tends to make the value at risk V_X smaller.

Exercise 6.8



See the attached **code*** for a solution using R.

a) i) $\mathrm{VaR}^{95}_{\mathrm{Ac\text{-}H}} = 5$.

ii) $\mathrm{VaR}^{80}_{\mathrm{Ac\text{-}H}} = 4$.

iii) $\mathrm{VaR}^{95}_{\mathrm{Pr\text{-}H}} = -3$.

iv) $\mathrm{VaR}^{80}_{\mathrm{Pr\text{-}H}} = -2$.

b) By Proposition 6.17, we have:

i) $\mathrm{VaR}^{95}_{\mathrm{Ac\text{-}G}} = 1.15{+}3.048 \times \mathrm{qnorm}(0.95){=}6.164$,

ii) $\mathrm{VaR}^{80}_{\mathrm{Ac\text{-}G}} = 1.15{+}3.048 \times \mathrm{qnorm}(0.80){=}3.71551$,

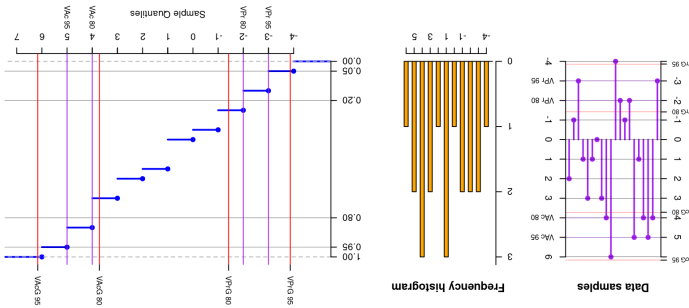
iii) $\mathrm{VaR}^{95}_{\mathrm{Pr\text{-}G}} = 1.15{-}3.048 \times \mathrm{qnorm}(0.95){=}{-}3.864$.

iv) $\mathrm{VaR}^{80}_{\mathrm{Pr\text{-}G}} = 1.15{-}3.048 \times \mathrm{qnorm}(0.80){=}{-}1.41551$,

Remark. The "Practitioner" Values at Risk can be better visualized after applying top-down and left-right symmetries (or a 180° rotation) to the original CDF, as in the next figure.

* Right-click to save as attachment (may not work on).

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Chapter 7

Exercise 7.1

a) Noting that $p = 1 - \mathrm{e}^{-\lambda \mathrm{VaR}_X^p}$ and using integration by parts on $\left[\mathrm{VaR}_X^p, \infty\right)$ with $u(x) = x$ and $v'(x) = \mathrm{e}^{-\lambda x}$, we have

$$\begin{aligned}
\mathbb{E}[X \mid X > \mathrm{VaR}_X^p] &= \frac{1}{\mathbb{P}(X > \mathrm{VaR}_X^p)} \int_{\mathrm{VaR}_X^p}^{\infty} x f_X(x) dx \\
&= \frac{\lambda}{1-p} \int_{\mathrm{VaR}_X^p}^{\infty} x \mathrm{e}^{-\lambda x} dx \\
&= \frac{\lambda}{1-p} \int_{\mathrm{VaR}_X^p}^{\infty} u(x) v'(x) dx \\
&= \frac{\lambda}{1-p} \left(\left[u(x) v(x) \right]_{\mathrm{VaR}_X^p}^{\infty} - \int_{\mathrm{VaR}_X^p}^{\infty} u'(x) v(x) dx \right) \\
&= \frac{\lambda}{1-p} \left(\left[-\frac{x}{\lambda} \mathrm{e}^{-\lambda x} \right]_{\mathrm{VaR}_X^p}^{\infty} + \frac{1}{\lambda} \int_{\mathrm{VaR}_X^p}^{\infty} \mathrm{e}^{-\lambda x} dx \right) \\
&= \frac{\lambda}{1-p} \left(\frac{\mathrm{VaR}_X^p}{\lambda} \mathrm{e}^{-\lambda \mathrm{VaR}_X^p} + \frac{1}{\lambda^2} \mathrm{e}^{-\lambda \mathrm{VaR}_X^p} \right) \\
&= \frac{\lambda}{1-p} \left(\frac{\mathrm{VaR}_X^p}{\lambda} (1-p) + \frac{1-p}{\lambda^2} \right) \\
&= \mathrm{VaR}_X^p + \frac{1}{\lambda} \\
&= \frac{1}{\lambda} - \frac{\log(1-p)}{\lambda}.
\end{aligned}$$

b) We have

$$\begin{aligned}\mathrm{TV}_X^p &= \frac{1}{1-p}\int_p^1 V_X^q dq\\ &= -\frac{1}{\lambda(1-p)}\int_p^1 \log(1-q)dq\\ &= -\frac{1}{\lambda(1-p)}\int_0^{1-p}(\log q)dq\\ &= \frac{1-p+(1-p)\log\dfrac{1}{1-p}}{\lambda(1-p)}\\ &= \frac{1}{\lambda}+\frac{1}{\lambda}\log\frac{1}{1-p}\\ &= \mathbb{E}[X]\left(1+\log\frac{1}{1-p}\right)\\ &= \mathbb{E}[X]+V_X^p.\end{aligned}$$

Exercise 7.2

a) We have

$$\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant z\}}\big]\geqslant\mathbb{E}\big[z\mathbb{1}_{\{X\geqslant z\}}\big]=z\mathbb{P}(X\geqslant z),$$

hence

$$\mathbb{E}[X\mid X\geqslant z]=\frac{\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant z\}}\big]}{\mathbb{P}(X\geqslant z)}\geqslant z. \tag{A.2}$$

Recall that $\mathbb{E}[X\mid X\geqslant z]$ is undefined if $\mathbb{P}(X\geqslant z)=0$.

b) Using (A.2), we have

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}\big[X\mathbb{1}_{\{X<z\}}\big]+\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant z\}}\big]\\ &\leqslant z\mathbb{E}\big[\mathbb{1}_{\{X<z\}}\big]+\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant z\}}\big]\\ &= z\mathbb{P}(X<z)+\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant z\}}\big]\\ &\leqslant \mathbb{E}[X\mid X\geqslant z]\mathbb{P}(X<z)+\mathbb{E}[X\mid X\geqslant z]\mathbb{P}(X\geqslant z)\\ &= \mathbb{E}[X\mid X\geqslant z].\end{aligned}$$

Note that $\mathbb{E}[X]=\mathbb{E}[X\mid X\geqslant z]$ when $\mathbb{P}(X<z)=0$, *i.e.* $\mathbb{P}(X\geqslant z)=1$.

c) If $\mathbb{P}(X>z)>0$ we have $\mathbb{E}\big[(X-z)\mathbb{1}_{\{X>z\}}\big]>0$, hence

$$\mathbb{E}\big[X\mathbb{1}_{\{X>z\}}\big]>\mathbb{E}\big[z\mathbb{1}_{\{X>z\}}\big]=z\mathbb{P}(X>z),$$

and

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$$\mathbb{E}[X \mid X > z] = \frac{\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big]}{\mathbb{P}(X > z)} > z. \tag{A.3}$$

Recall that $\mathbb{E}[X \mid X > z]$ is undefined if $\mathbb{P}(X > z) = 0$.

d) Using (A.3), we have

$$\begin{aligned}
\mathbb{E}[X] &= \mathbb{E}\big[X \mathbb{1}_{\{X\leqslant z\}}\big] + \mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big] \\
&\leqslant z\mathbb{E}\big[\mathbb{1}_{\{X\leqslant z\}}\big] + \mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big] \\
&= z\mathbb{P}(X \leqslant z) + \mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big] \\
&\leqslant \mathbb{E}[X \mid X > z]\mathbb{P}(X \leqslant z) + \mathbb{E}[X \mid X > z]\mathbb{P}(X > z) \\
&= \mathbb{E}[X \mid X > z].
\end{aligned}$$

Note that $\mathbb{E}[X] = \mathbb{E}[X \mid X > z]$ when $\mathbb{P}(X \leqslant z) = 0$, *i.e.* $\mathbb{P}(X > z) = 1$.

e) We have $\mathbb{P}(X \geqslant z) = \mathbb{P}(X > z)$, and

$$\begin{aligned}
\mathbb{E}[X \mid X > z] &= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big]}{\mathbb{P}(X > z)} \\
&= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X\geqslant z\}}\big] - z\mathbb{P}(X = z)}{\mathbb{P}(X \geqslant z)} \\
&= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X\geqslant z\}}\big]}{\mathbb{P}(X \geqslant z)} \\
&= \mathbb{E}[X \mid X > z].
\end{aligned}$$

f) Using (A.3), we have

$$\begin{aligned}
\mathbb{E}[X \mid X \geqslant z] &= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X\geqslant z\}}\big]}{\mathbb{P}(X \geqslant z)} \\
&= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big] + z\mathbb{P}(X = z)}{\mathbb{P}(X > z) + \mathbb{P}(X = z)} \\
&< \frac{\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big] + \mathbb{P}(X = z)\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big]/\mathbb{P}(X > z)}{\mathbb{P}(X > z) + \mathbb{P}(X = z)} \\
&= \frac{\mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big]}{\mathbb{P}(X > z)} \\
&= \mathbb{E}[X \mid X > z].
\end{aligned}$$

g) When $\mathbb{P}(X \leqslant z) > 0$, from (A.3) we find

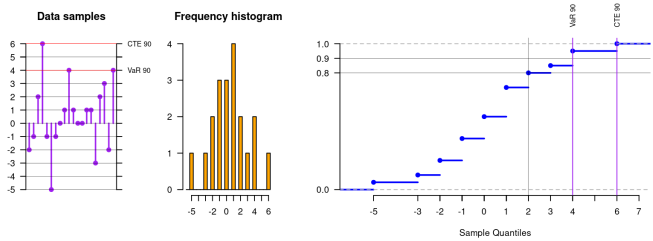
$$\mathbb{E}[X] \leqslant z\mathbb{P}(X \leqslant z) + \mathbb{E}\big[X \mathbb{1}_{\{X>z\}}\big]$$

$$< \mathbb{E}[X \mid X > z]\mathbb{P}(X \leqslant z) + \mathbb{E}[X \mid X > z]\mathbb{P}(X > z)$$
$$= \mathbb{E}[X \mid X > z].$$

h) By (6.10) we have $\mathbb{P}\big(X \leqslant V_X^p\big) \geqslant p > 0$, hence by (g) above we find $\mathrm{CTE}_X^p = \mathbb{E}\big[X \,\big|\, X > V_X^p\big] > \mathbb{E}[X]$.

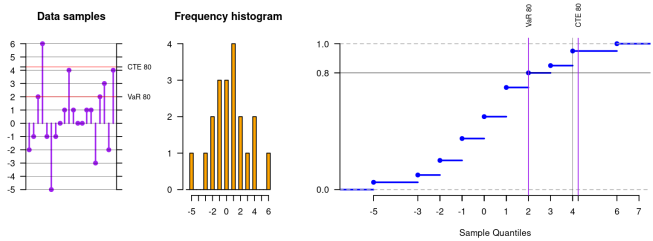
Exercise 7.3

a) We have $\mathrm{VaR}_X^{0.9} = 4$ and $\mathrm{CTE}_X^{0.9} = 6$.



b) We have $\mathrm{VaR}_X^{0.8} = 2$ and

$$\mathrm{CTE}_X^{0.8} = \frac{3 + 2 \times 4 + 6}{4} = \frac{17}{4} = 4.25.$$



Equivalently, we have

$$\mathrm{CTE}_X^{0.8} = \frac{0.05 \times 3 + 0.1 \times 4 + 0.05 \times 6}{0.05 + 0.1 + 0.05}$$
$$= \frac{0.05 \times 3 + 0.1 \times 4 + 0.05 \times 6}{0.2}$$

$$= \frac{0.85}{0.2} = 4.25.$$

Exercise 7.4

a) $\mathrm{VaR}_X^{90\%} = 4$.

b) $\mathbb{E}\big[X\mathbb{1}_{\{X>V_X^{90\%}\}}\big] = \dfrac{5+6}{23} = \dfrac{11}{23}$.

c) $\mathbb{P}\big(X > V_X^{90\%}\big) = \dfrac{2}{23}$.

d) $\mathrm{CTE}_X^{90\%} = \mathbb{E}\big[X \mid X > V_X^{90\%}\big] = \dfrac{\mathbb{E}\big[X\mathbb{1}_{\{X>V_X^{90\%}\}}\big]}{\mathbb{P}\big(X > V_X^{90\%}\big)} = \dfrac{5+6}{2} = \dfrac{11}{2} = 5.50.$

e) $\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant V_X^{90\%}\}}\big] = \dfrac{4+5+6}{23} = \dfrac{15}{23}$.

f) $\mathbb{P}\big(X \geqslant V_X^{90\%}\big) = \dfrac{3}{23}$.

g) $\mathrm{ES}_X^{90\%} = \dfrac{1}{1-p}\big(\mathbb{E}\big[X\mathbb{1}_{\{X\geqslant V_X^{90\%}\}}\big] + V_X^{90\%}\big(1-p-\mathbb{P}\big(X \geqslant V_X^{90\%}\big)\big)\big) = 10\times \dfrac{4+5+6}{23} + 10\times 4\left(0.1 - \dfrac{3}{23}\right) = \dfrac{150}{23} + 40\times\dfrac{2.3-3}{23} = \dfrac{150-40\times 0.7}{23} = \dfrac{122}{23} = 5.304.$

h) $\mathrm{TV}_X^{90\%} = \dfrac{1}{1-p}\displaystyle\int_p^1 V_X^q dq = \dfrac{1}{1-p}\left(\int_p^{21/23} V_X^q dq + \int_{21/23}^{22/23} V_X^q dq + \int_{22/23}^1 V_X^q dq\right)$
$= \dfrac{1}{1-p}\left(\displaystyle\int_p^{21/23} 4dq + \int_{21/23}^{22/23} 5dq + \int_{22/23}^1 6dq\right)$
$= \dfrac{1}{1-p}\left(4\left(\dfrac{21}{23} - p\right) + \dfrac{5}{23} + \dfrac{6}{23}\right) = \dfrac{84-92p+5+6}{23(1-p)} = \dfrac{122}{23} = 5.304.$

We note that $\mathrm{ES}_X^{90\%} = \mathrm{TV}_X^{90\%}$ according to Proposition 7.13. The attached **code** computes the above risk measures, as illustrated in Figure S.11.

```
> source("var-cte_quiz.R")
VaR90= 4, Threshold= 0.9130435
CTE90= 5.5
ES90= 5.304348
```

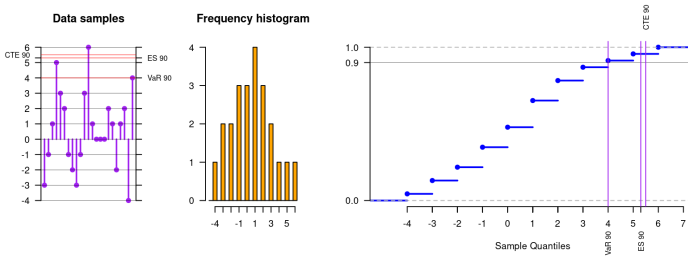


Fig. S.11: Value at Risk and Expected Shortfall for small data.

Exercise 7.5

a) The Value at Risk is $V_X^{98\%} = 100$.

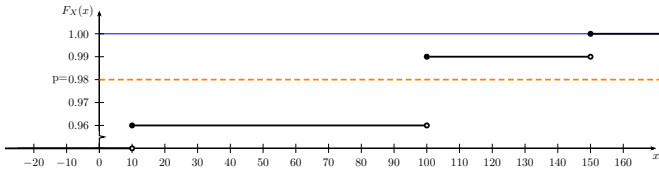


Fig. S.12: Cumulative distribution function of X.

b) Taking $p = 0.98$, we have

$$\begin{aligned}\mathrm{TV}_X^{98\%} &= \frac{1}{1-p}\int_p^1 V_X^q dq \\ &= \frac{1}{0.02}\left((0.99-0.98)\times 100 + (1-0.99)\times 150\right) = 125.\end{aligned}$$

c) We have

$$\begin{aligned}\mathrm{CTE}_X^{98\%} &= \frac{1}{\mathbb{P}(X > V_X^p)}\mathbb{E}\left[X\mathbb{1}_{\{X>V_X^p\}}\right] \\ &= \frac{1}{0.01}\times 150\times 0.01 = 150.\end{aligned}$$

d) We have

$$\mathrm{ES}_X^{98\%} = \frac{1}{1-p}\mathbb{E}\left[X\mathbb{1}_{\{X\geqslant V_X^p\}}\right] + \frac{V_X^p}{1-p}\left(1-p-\mathbb{P}(X\geqslant V_X)\right)$$

$$= \frac{1}{0.02}(100 \times 0.03 + 150 \times 0.01) + \frac{100}{0.02}(0.02 - (0.03 + 0.01))$$
$$= \frac{4.5}{0.02} + \frac{100}{0.02}(0.02 - (0.03 + 0.01)) = 125.$$

Note that we also have

$$\begin{aligned} \mathrm{ES}_X^{98\%} &= \frac{1}{1-p}\mathbb{E}\left[X\mathbb{1}_{\{X>V_X^p\}}\right] + \frac{V_X^p}{1-p}(1-p-\mathbb{P}(X>V_X)) \\ &= \frac{1}{0.02}(150 \times 0.01) + \frac{100}{0.02}(0.02 - 0.01) \\ &= 125, \end{aligned}$$

hence the Expected Shortfall $\mathrm{ES}_X^{98\%}$ does coincide with the tail value at risk $TV_X^{98\%}$.

Exercise 7.6

a) The cumulative distribution function of X is given by the following graph:

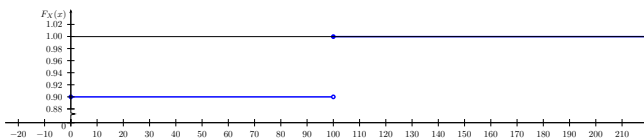


Fig. S.13: Cumulative distribution function of X.

b) The distribution of $X+Y$ is given by

$$\mathbb{P}(X+Y=0) = 81\%, \quad \mathbb{P}(X+Y=100) = 18\%, \quad \mathbb{P}(X+Y=200) = 1\%.$$

The cumulative distribution function of $X+Y$ is given by the following graph:

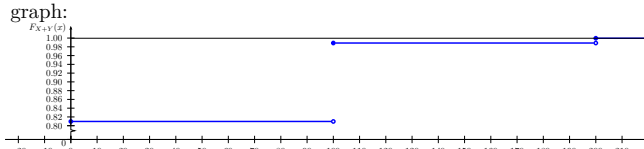


Fig. S.14: Cumulative distribution function of $X+Y$.

c) We have $V_{X+Y}^{99\%} = V_{X+Y}^{95\%} = V_{X+Y}^{90\%} = 100$.

Note that we have $V_{X+Y}^{99\%} = 100$ because

$$V_X^{99\%} = \inf\{x \in \mathbb{R} \ : \ \mathbb{P}(X \leqslant x) \geqslant 0.99\} = 100.$$

d) We have

$$\mathrm{TV}_X^{90\%} = \frac{1}{1-0.9}\int_{0.9}^1 V_X^q dq = 100 \times \frac{1-0.9}{1-0.9} = 100.$$

e) We have

$$\begin{aligned}\mathrm{TV}_{X+Y}^{99\%} &= \frac{1}{1-0.9}\int_{0.9}^1 V_{X+Y}^q dq \\ &= \frac{1}{0.1}\left(\int_{0.9}^{0.99} 100 dq + \int_{0.99}^1 200 dq\right) \\ &= \frac{1}{0.1}\left(100 \times 0.09 + 200 \times 0.01\right) \\ &= 110,\end{aligned}$$

and

$$\begin{aligned}\mathrm{TV}_{X+Y}^{80\%} &= \frac{1}{1-0.8}\int_{0.9}^1 V_{X+Y}^q dq \\ &= \frac{1}{0.2}\left(\int_{0.8}^{0.81} 0 dq + \int_{0.81}^{0.99} 100 dq + \int_{0.99}^1 200 dq\right) \\ &= \frac{1}{0.2}\left(100 \times 0.18 + 200 \times 0.01\right) = 100.\end{aligned}$$

Exercise 7.7 (Exercise 6.3 continued).

a) For all $p \in [0.99, 1]$ we have

$$\begin{aligned}\mathrm{TV}_X^p &= \frac{1}{1-p}\int_p^1 V_X^q dq \\ &= \frac{1}{1-p}\int_p^1 (5000q - 4850) dq \\ &= \frac{1}{1-p}\left(5000\frac{(1-p^2)}{2} - (1-p)4850\right) \\ &= 2500p - 2350.\end{aligned}$$

In particular,

$$\mathrm{TV}_X^{99\%} = 2500 \times 0.99 - 2350 = 125 \geqslant V_X^{99\%} = 100.$$

b) We have $V_X^{98\%} = 100$ and

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$$\begin{aligned}
\mathrm{CTE}_X^{98\%} &= \mathbb{E}\big[X \mid X > V_X^{98\%}\big] \\
&= \frac{1}{\mathbb{P}\big(X > V_X^{98\%}\big)}\mathbb{E}\left[X\mathbb{1}_{\{X>V_X^{98\%}\}}\right] \\
&= \frac{1}{0.01}\int_{100}^{\infty} x f_X(x)dx \\
&= \frac{1}{0.01}\int_{100}^{\infty} x\frac{dF_X(x)}{dx}dx \\
&= \frac{1}{0.01}\frac{0.01}{50}\int_{100}^{150} x dx \\
&= \frac{150^2-100^2}{2\times 50} \\
&= 125.
\end{aligned}$$

Note that

$$\begin{aligned}
\mathrm{TV}_X^{98\%} &= \frac{0.01}{0.02}\times 100 + \frac{1}{0.02}\left(5000\frac{(1-0.99^2)}{2} - 0.01\times 4850\right) \\
&= 112.50 \\
&\geqslant V_X^{98\%} = 100,
\end{aligned}$$

which differs from $\mathrm{CTE}_X^{98\%} = 125$ since

$$\mathbb{P}\big(X = V_X^{98\%}\big) = \mathbb{P}(X = 100) = 0.02 > 0.$$

Exercise 7.8

a) We have

$$V_X^p := \inf\big\{x \in \mathbb{R} \ : \ \mathbb{P}(X \leqslant x) \geqslant p\big\} = \log\frac{p}{1-p}.$$

b) We have

$$\begin{aligned}
\mathbb{E}[X \mid X > \mathrm{VaR}_X^p] &= \frac{1}{\mathbb{P}(X > \mathrm{VaR}_X^p)}\int_{\mathrm{VaR}_X^p}^{\infty} x f_X(x)dx \\
&= \frac{1}{1-p}\int_{\mathrm{VaR}_X^p}^{\infty} x\mathrm{e}^{-\lambda x}dx \\
&= \frac{1}{1-p}\int_{\mathrm{VaR}_X^p}^{\infty} \frac{x\mathrm{e}^{-x}}{(1+\mathrm{e}^{-x})^2}dx \\
&= \frac{1}{1-p}\log(1+\mathrm{e}^{\mathrm{VaR}_X^p}) - \frac{\mathrm{VaR}_X^p\,\mathrm{e}^{\mathrm{VaR}_X^p}}{1+\mathrm{e}^{\mathrm{VaR}_X^p}}
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{1-p}\log\left(1+\frac{p}{1-p}\right) - \frac{1}{1-p}\frac{p}{1-p}\frac{1}{1+\frac{p}{1-p}}\log\frac{p}{1-p}\\
&= \frac{1}{1-p}\log\frac{1}{1-p} - \frac{p}{1-p}\log\frac{p}{1-p}\\
&= -\frac{p}{1-p}\log p - \log(1-p).
\end{aligned}
$$

c) We have

$$
\begin{aligned}
\mathrm{TV}_X^p &= \frac{1}{1-p}\int_p^1 V_X^q dq\\
&= \frac{1}{1-p}\int_p^1 \log\frac{q}{1-q}dq\\
&= \frac{1}{1-p}\int_p^1 \log q dq - \frac{1}{1-p}\int_p^1 \log(1-q)dq\\
&= \frac{1}{1-p}\int_p^1 \log q dq - \frac{1}{1-p}\int_0^{1-p} \log q dq\\
&= \frac{1}{1-p}\int_p^1 \log q dq - \frac{1}{1-p}\left(\int_0^1 \log q dq - \int_{1-p}^1 \log q dq\right)\\
&= \frac{p-1-p\log p}{1-p} - \frac{-1+p+(1-p)\log(1-p)}{1-p}\\
&= -\frac{p}{1-p}\log p - \log(1-p).
\end{aligned}
$$

Exercise 7.9

a) We have

$$
q\mathbb{P}(Z \geqslant q) = \mathbb{E}\big[q\mathbb{1}_{\{Z\geqslant q\}}\big] \leqslant \mathbb{E}\big[Z\mathbb{1}_{\{Z\geqslant q\}}\big] = \int_q^\infty x f_Z(x)dx, \qquad q \geqslant 0.
$$

b) We have

$$
\begin{aligned}
\int_q^\infty x f_Z(x)dx &= \int_q^\infty x\phi(x)dx\\
&= \frac{1}{\sqrt{2\pi}}\int_q^\infty x\mathrm{e}^{-x^2/2}dx\\
&= -\frac{1}{\sqrt{2\pi}}\Big[\mathrm{e}^{-x^2/2}\Big]_q^\infty\\
&= \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-q^2/2}\\
&= \phi(q), \qquad q \geqslant 0,
\end{aligned}
$$

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and $1-p=\mathbb{P}(Z\geqslant q)$, hence

$$(1-p)q\leqslant\int_q^\infty xf_Z(x)dx=\phi(q),\qquad q\geqslant 0.$$

c) Taking $q:=q_Z^p$ with $1-p=\mathbb{P}(Z\geqslant q_Z^p)$, we recover

$$V_X^p=\mu_X+\sigma_X q_Z^p\leqslant\mu_X+\frac{\sigma_X}{1-p}\phi(q_Z^p)=\mathrm{CTE}_X^p,$$

see Proposition 7.5.

Chapter 8

Exercise 8.1 By differentiation of (8.3), *i.e.*

$$\begin{aligned}\mathbb{P}(\tau<T\mid\mathcal{F}_t)&:=\mathbb{P}(S_T<K\mid\mathcal{F}_t)\\&=\Phi\left(-\frac{(\mu-\sigma^2/2)(T-t)+\log(S_t/K))}{\sigma\sqrt{T-t}}\right),\qquad T\geqslant t,\end{aligned}$$

with respect to T, we find

$$\begin{aligned}\mathrm{d}\mathbb{P}(\tau\leqslant T\mid\mathcal{F}_t)=&\frac{dT}{2\sigma\sqrt{2\pi(T-t)}}\left(\frac{\sigma^2}{2}-\mu+\frac{\log(S_t/K)}{T-t}\right)\\&\times\exp\left(-\frac{\big((\mu-\sigma^2/2))(T-t)+\log(S_t/K))\big)^2}{2(T-t)\sigma^2}\right),\end{aligned}$$

provided that $\mu<\sigma^2/2$.

Exercise 8.2 Consider the first hitting time

$$\tau_K:=\inf\{u\geqslant t\ :\ S_u\leqslant K\}$$

of the level $K>0$ starting from $S_t>K$. By Lemma 15.1 in Privault (2022), we have

$$\mathbb{E}^*\big[\mathrm{e}^{-(\tau_K-t)r}\,\big|\,\mathcal{F}_t\big]=\left(\frac{K}{S_t}\right)^{2r/\sigma^2},$$

provided that $S_t\geqslant K$.

Exercise 8.3

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a) We have

$$\begin{aligned}
\mathbb{E}[X_k X_l] &= \mathbb{E}\left[(a_k M + \sqrt{1-a_k^2}Z_k)(a_l M + \sqrt{1-a_l^2}Z_l)\right] \\
&= \mathbb{E}\left[a_k a_l M^2 + a_k M\sqrt{1-a_l^2}Z_l + a_l M\sqrt{1-a_k^2}Z_k + \sqrt{1-a_k^2}\sqrt{1-a_l^2}Z_k Z_l\right] \\
&= a_k a_l \mathbb{E}\big[M^2\big] + a_k\sqrt{1-a_l^2}\mathbb{E}[Z_l M] + a_l\sqrt{1-a_k^2}\mathbb{E}[Z_k M] \\
&\quad + \sqrt{1-a_k^2}\sqrt{1-a_l^2}\mathbb{E}[Z_k Z_l] \\
&= a_k a_l \mathbb{E}\big[M^2\big] + a_k\sqrt{1-a_l^2}\mathbb{E}[Z_l]\mathbb{E}[M] + a_l\sqrt{1-a_k^2}\mathbb{E}[Z_k]\mathbb{E}[M] \\
&\quad + \sqrt{1-a_k^2}\sqrt{1-a_l^2}\mathbb{1}_{\{k=l\}} \\
&= a_k a_l + (1-a_k^2)\mathbb{1}_{\{k=l\}} \\
&= \mathbb{1}_{\{k=l\}} + a_k a_l \mathbb{1}_{\{k\neq l\}}, \qquad k,l = 1,2,\ldots,n,
\end{aligned}$$

b) We check that the vector $(X_1,\ldots,X_n)$, with covariance matrix (8.13) has the probability density function

$$\begin{aligned}
&\varphi(x_1,\ldots,x_n) \\
&\quad = \frac{1}{(2\pi)^{n/2}}\prod_{k=1}^n (1-a_k^2)^{-1/2}\int_{-\infty}^{\infty} \mathrm{e}^{-\frac{(x_1-a_1 m)^2}{2(1-a_1^2)}}\cdots \mathrm{e}^{-\frac{(x_n-a_n m)^2}{2(1-a_n^2)}}\frac{\mathrm{e}^{-m^2/2}}{\sqrt{2\pi}}dm
\end{aligned}$$

which is jointly Gaussian, with marginals given by

$$\begin{aligned}
x_k &\longmapsto \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty}\varphi(x_1,\ldots,x_n)dx_1\cdots dx_{k-1}dx_{k+1}\cdots dx_n \\
&= \frac{1}{\sqrt{2\pi(1-a_k^2)}}\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{(x_k-a_k m)^2}{2(1-a_k^2)}}\frac{\mathrm{e}^{-m^2/2}}{\sqrt{2\pi}}dm \\
&= \frac{1}{2\pi\sqrt{1-a_k^2}}\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{(x_k-a_k m)^2}{2(1-a_k^2)}-m^2/2}dm \\
&= \frac{1}{2\pi\sqrt{1-a_k^2}}\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{x_k^2-2a_k x_k m+m^2}{2(1-a_k^2)}}dm \\
&= \frac{\mathrm{e}^{-x_k^2/2}}{2\pi\sqrt{1-a_k^2}}\int_{-\infty}^{\infty}\mathrm{e}^{-\frac{(m-a_k x_k)^2}{2(1-a_k^2)}}dm
\end{aligned}$$

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$$= \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-x_k^2/2}, \qquad x_k \in \mathbb{R}.$$

c) We have

$$\begin{aligned}
\varphi(x_1,\ldots,x_n) &= \frac{1}{(2\pi)^{n/2}} \prod_{k=1}^{n} (1-a_k^2)^{-1/2} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{(x_1 - a_1 m)^2}{2(1-a_1^2)}} \cdots \mathrm{e}^{-\frac{(x_n - a_n m)^2}{2(1-a_n^2)}} \frac{\mathrm{e}^{-m^2/2}}{\sqrt{2\pi}} dm \\
&= \frac{1}{(2\pi)^{n/2}} \prod_{k=1}^{n} (1-a_k^2)^{-1/2} \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2 + a_1^2 m^2 - 2x_1 a_1 m}{1-a_1^2} + \cdots + \frac{x_n^2 + a_n^2 m^2 - 2x_n a_n m}{1-a_n^2} + m^2 \right)} \frac{dm}{\sqrt{2\pi}} \\
&= \frac{1}{(2\pi)^{n/2}\sqrt{2\pi}} \mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2}{1-a_1^2} + \cdots + \frac{x_n^2}{1-a_n^2} \right)} \prod_{k=1}^{n} (1-a_k^2)^{-1/2} \\
&\quad \int_{-\infty}^{\infty} \mathrm{e}^{-\frac{m^2}{2}\left(1 + \frac{a_1^2}{1-a_1^2} + \cdots + \frac{a_n^2}{1-a_n^2} \right) + 2m \left(\frac{x_1 a_1}{2(1-a_1^2)} + \cdots + \frac{x_n a_n}{2(1-a_n^2)} \right)} dm \\
&= \frac{\mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2}{1-a_1^2} + \cdots + \frac{x_n^2}{1-a_n^2} \right)}}{\sqrt{(2\pi)^n (1-a_1^2)\cdots(1-a_n^2)}} \exp\left(\frac{\frac{1}{2}\left(\frac{x_1 a_1}{1-a_1^2} + \cdots + \frac{x_n a_n}{1-a_n^2} \right)^2}{1 + \frac{a_1^2}{1-a_1^2} + \cdots + \frac{a_n^2}{1-a_n^2}} \right) \\
&\quad \times \left(1 + \frac{a_1^2}{1-a_1^2} + \cdots + \frac{a_n^2}{1-a_n^2} \right)^{-1/2} \\
&= \frac{\mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2}{1-a_1^2} + \cdots + \frac{x_n^2}{1-a_n^2} \right)}}{\sqrt{(2\pi)^n \alpha^2 (1-a_1^2)\cdots(1-a_n^2)}} \exp\left(\frac{1}{2\alpha^2} \left(\frac{x_1 a_1}{1-a_1^2} + \cdots + \frac{x_n a_n}{1-a_n^2} \right)^2 \right) \\
&= \frac{\mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2}{1-a_1^2} + \cdots + \frac{x_n^2}{1-a_n^2} \right)}}{\sqrt{(2\pi)^n \alpha^2 (1-a_1^2)\cdots(1-a_n^2)}} \exp\left(\frac{1}{2\alpha^2} \left(\frac{x_1 a_1}{1-a_1^2} + \cdots + \frac{x_n a_n}{1-a_n^2} \right)^2 \right) \\
&= \frac{\mathrm{e}^{-\frac{1}{2}\left(\frac{x_1^2}{1-a_1^2}\left(1 - \frac{a_1^2}{\alpha^2(1-a_1^2)}\right) + \cdots + \frac{x_n^2}{1-a_n^2}\left(1 - \frac{a_n^2}{\alpha^2(1-a_n^2)}\right) \right)}}{\sqrt{(2\pi)^n \alpha^2 (1-a_1^2)\cdots(1-a_n^2)}} \exp\left(\frac{1}{2\alpha^2} \sum_{1 \leqslant p \neq l \leqslant n} \frac{x_p x_l a_p a_l}{(1-a_p^2)(1-a_l^2)} \right) \\
&= \frac{1}{\sqrt{(2\pi)^n \det \Sigma}} \mathrm{e}^{-\frac{1}{2}\langle x, \Sigma^{-1} x \rangle},
\end{aligned}$$

where

$$\alpha^2 := 1 + \frac{a_1^2}{1-a_1^2} + \cdots + \frac{a_n^2}{1-a_n^2},$$

and

$$\Sigma^{-1} = \frac{1}{\alpha^2} \begin{bmatrix} \frac{\alpha^2(1-a_1^2)-a_1^2}{(1-a_1^2)^2} & \frac{-a_1a_2}{(1-a_1^2)(1-a_2^2)} & \cdots & \frac{-a_1a_n}{(1-a_1^2)(1-a_n^2)} \\ \frac{-a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2(1-a_2^2)-a_2^2}{(1-a_2^2)^2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \frac{\alpha^2(1-a_{n-1}^2)-a_{n-1}^2}{(1-a_{n-1}^4)} & \frac{-a_{n-1}a_n}{(1-a_{n-1}^2)(1-a_n^2)} \\ \frac{-a_na_1}{(1-a_n^2)(1-a_1^2)} & \ddots & \frac{-a_na_{n-1}}{(1-a_n^2)(1-a_{n-1}^2)} & \frac{\alpha^2(1-a_n^2)-a_n^2}{(1-a_n^2)^2} \end{bmatrix}.$$

Exercise 8.4 We have

$$\Sigma = \begin{bmatrix} 1 & a_1a_2 \\ a_2a_1 & 1 \end{bmatrix},$$

and letting

$$\begin{aligned} \alpha^2 &:= 1 + \frac{a_1^2}{1-a_1^2} + \frac{a_2^2}{1-a_2^2} \\ &= \frac{(1-a_1^2)(1-a_2^2) + a_1^2(1-a_2^2) + a_2^2(1-a_1^2)}{(1-a_1^2)(1-a_2^2)} \\ &= \frac{1-a_2^2a_1^2}{(1-a_1^2)(1-a_2^2)}, \end{aligned}$$

we find

$$\begin{aligned} \Sigma^{-1} &= \frac{1}{\alpha^2} \begin{bmatrix} \frac{\alpha^2(1-a_1^2)-a_1^2}{(1-a_1^2)^2} & -\frac{a_1a_2}{(1-a_1^2)(1-a_2^2)} \\ -\frac{a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2(1-a_2^2)-a_2^2}{(1-a_2^2)^2} \end{bmatrix} \\ &= \frac{1}{\alpha^2} \begin{bmatrix} \frac{\alpha^2}{1-a_1^2}\left(1 - \frac{(1-a_2^2)a_1^2}{1-a_2^2a_1^2}\right) & -\frac{a_1a_2}{(1-a_1^2)(1-a_2^2)} \\ -\frac{a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2}{1-a_2^2}\left(1 - \frac{(1-a_1^2)a_2^2}{1-a_2^2a_1^2}\right) \end{bmatrix} \\ &= \frac{1}{\alpha^2} \begin{bmatrix} \frac{\alpha^2}{1-a_2^2a_1^2} & -\frac{a_1a_2}{(1-a_1^2)(1-a_2^2)} \\ -\frac{a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2}{1-a_2^2a_1^2} \end{bmatrix} \\ &= \frac{(1-a_1^2)(1-a_2^2)}{1-a_2^2a_1^2} \begin{bmatrix} \frac{\alpha^2}{1-a_2^2a_1^2} & -\frac{a_1a_2}{(1-a_1^2)(1-a_2^2)} \\ -\frac{a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2}{1-a_2^2a_1^2} \end{bmatrix} \\ &= \frac{1}{\alpha^2} \begin{bmatrix} \frac{\alpha^2}{1-a_2^2a_1^2} & -\frac{a_1a_2}{(1-a_1^2)(1-a_2^2)} \\ -\frac{a_2a_1}{(1-a_2^2)(1-a_1^2)} & \frac{\alpha^2}{1-a_2^2a_1^2} \end{bmatrix} \end{aligned}$$

$$= \frac{1}{1-a_2^2 a_1^2}\begin{bmatrix} 1 & -a_1 a_2 \\ -a_1 a_2 & 1 \end{bmatrix}.$$

In particular, the case $n = 2$ is able to recover all two-dimensional copulas by setting the correlation coefficient $\rho = a_1 a_2$. In the general case, Σ is parametrized by n numbers, which offers less degrees of freedom compared with the joint Gaussian copula correlation method which relies on $n(n-1)/2$ coefficients, see also Exercise 8.3.

Chapter 9

Exercise 9.1 By absence of arbitrage we have $(1-\alpha)\mathrm{e}^{r_d T} = \mathrm{e}^{rT}$, hence $\alpha = 1 - \mathrm{e}^{(r-r_d)T}$.

Exercise 9.2 We have

$$\begin{aligned}
\lambda &= \frac{1}{T-t}\log\frac{P(t,T)}{P_d(t,T)} \\
&= \frac{1}{T-t}\log\frac{\mathrm{e}^{-(T-t)r}}{P_d(t,T)} \\
&= -r - \frac{1}{2}\log P_d(t,T) \\
&= -0.025 - \frac{1}{2}\log 0.935 \\
&= 0.008604375 \\
&= 0.86\%,
\end{aligned}$$

hence the default probability is

$$p_D = 1 - \mathrm{e}^{-(T-t)\lambda} = 1 - \mathrm{e}^{-2\times 0.008604375} = 0.01706153 = 1.71\%.$$

Exercise 9.3

a) The bond payoff $\mathbb{1}_{\{\tau>T-t\}}$ is discounted according to the risk-free rate, before taking expectation.
b) We have $\mathbb{E}\left[\mathbb{1}_{\{\tau>T-t\}}\right] = \mathrm{e}^{-\lambda(T-t)}$, hence $P_d(t,T) = \mathrm{e}^{-(\lambda+r)(T-t)}$.
c) We have $P_M(t,T) = \mathrm{e}^{-(\lambda+r)(T-t)}$, hence $\lambda = -r - \dfrac{1}{T-t}\log P_M(t,T)$.

Exercise 9.4

a) We have

$$r_t = -a\int_0^t r_s ds + \sigma B_t^{(1)}, \quad t \geqslant 0,$$

hence

$$\begin{aligned}\int_0^t r_s ds &= \frac{1}{a}\big(\sigma B_t^{(1)} - r_t\big)\\ &= \frac{\sigma}{a}\left(B_t^{(1)} - \int_0^t \mathrm{e}^{-(t-s)a} dB_s^{(1)}\right)\\ &= \frac{\sigma}{a}\int_0^t (1-\mathrm{e}^{-(t-s)a}) dB_s^{(1)},\end{aligned}$$

and

$$\begin{aligned}&\int_t^T r_s ds = \int_0^T r_s ds - \int_0^t r_s ds\\ &= \frac{\sigma}{a}\int_0^T (1-\mathrm{e}^{-(T-s)a}) dB_s^{(1)} - \frac{\sigma}{a}\int_0^t (1-\mathrm{e}^{-(t-s)a}) dB_s^{(1)}\\ &= -\frac{\sigma}{a}\left(\int_0^t (\mathrm{e}^{-(T-s)a} - \mathrm{e}^{-(t-s)a}) dB_s^{(1)} + \int_t^T (\mathrm{e}^{-(T-s)a} - 1) dB_s^{(1)}\right)\\ &= -\frac{\sigma}{a}(\mathrm{e}^{-(T-t)a} - 1)\int_0^t \mathrm{e}^{-(t-s)a} dB_s^{(1)} - \frac{\sigma}{a}\int_t^T (\mathrm{e}^{-(T-s)a} - 1) dB_s^{(1)}\\ &= -\frac{1}{a}(\mathrm{e}^{-(T-t)a} - 1) r_t - \frac{\sigma}{a}\int_t^T (\mathrm{e}^{-(T-s)a} - 1) dB_s^{(1)}.\end{aligned}$$

The answer for λ_t is similar.

b) As a consequence of the answer to the previous question, we have

$$\mathbb{E}\left[\int_t^T r_s ds + \int_t^T \lambda_s ds \;\middle|\; \mathcal{F}_t\right] = C(a,t,T) r_t + C(b,t,T)\lambda_t,$$

and

$$\begin{aligned}&\mathrm{Var}\left[\int_t^T r_s ds + \int_t^T \lambda_s ds \;\middle|\; \mathcal{F}_t\right]\\ &= \mathrm{Var}\left[\int_t^T r_s ds \;\middle|\; \mathcal{F}_t\right] + \mathrm{Var}\left[\int_t^T \lambda_s ds \;\middle|\; \mathcal{F}_t\right]\\ &\quad + 2\,\mathrm{Cov}\left(\int_t^T X_s ds, \int_t^T Y_s ds \;\middle|\; \mathcal{F}_t\right)\\ &= \frac{\sigma^2}{a^2}\int_t^T (\mathrm{e}^{-(T-s)a} - 1)^2 ds\\ &\quad + 2\rho\frac{\sigma\eta}{ab}\int_t^T (\mathrm{e}^{-(T-s)a} - 1)(\mathrm{e}^{-(T-s)b} - 1) ds\end{aligned}$$

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$$
\begin{aligned}
&+\frac{\eta^2}{b^2}\int_t^T (\mathrm{e}^{-(T-s)b}-1)^2 ds\\
&=\sigma^2\int_t^T C^2(a,s,T)ds+2\rho\sigma\eta\int_t^T C(a,s,T)C(b,s,T)ds\\
&\quad+\eta^2\int_t^T C^2(b,sT)ds,
\end{aligned}
$$

from the Itô isometry.

Exercise 9.5 (Exercise 9.4 continued).

a) We use the fact that $(r_t,\lambda_t)_{t\in[0,T]}$ is a Markov process.
b) We use the tower property (A.33) of the conditional expectation given $\mathcal{F}_t$.

c) Writing $F(t,r_t,\lambda_t)=P(t,T)$, we have

$$
\begin{aligned}
&d\left(\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}P(t,T)\right)\\
&=-(r_t+\lambda_t)\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}P(t,T)dt+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}dP(t,T)\\
&=-(r_t+\lambda_t)\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}P(t,T)dt+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}dF(t,r_t,\lambda_t)\\
&=-(r_t+\lambda_t)\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}P(t,T)dt+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial F}{\partial x}(t,r_t,\lambda_t)dr_t\\
&+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial F}{\partial y}(t,r_t,\lambda_t)d\lambda_t+\frac{1}{2}\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial^2 F}{\partial x^2}(t,r_t,\lambda_t)\sigma_1^2(t,r_t)dt\\
&+\frac{1}{2}\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial^2 F}{\partial y^2}(t,r_t,\lambda_t)\sigma_2^2(t,\lambda_t)dt\\
&+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\rho\frac{\partial^2 F}{\partial x\partial y}(t,r_t,\lambda_t)\sigma_1(t,r_t)\sigma_2(t,\lambda_t)dt+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial F}{\partial t}(t,r_t,\lambda_t)dt\\
&=\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial F}{\partial x}(t,r_t,\lambda_t)\sigma_1(t,r_t)dB_t^{(1)}+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\frac{\partial F}{\partial y}(t,r_t,\lambda_t)\sigma_2(t,\lambda_t)dB_t^{(2)}\\
&+\mathrm{e}^{-\int_0^t(r_s+\lambda_s)ds}\left(-(r_t+\lambda_t)P(t,T)+\frac{\partial F}{\partial x}(t,r_t,\lambda_t)\mu_1(t,r_t)\right.\\
&+\frac{\partial F}{\partial y}(t,r_t,\lambda_t)\mu_2(t,\lambda_t)+\frac{1}{2}\frac{\partial^2 F}{\partial x^2}(t,r_t,\lambda_t)\sigma_1^2(t,r_t)+\frac{1}{2}\frac{\partial^2 F}{\partial y^2}(t,r_t,\lambda_t)\sigma_2^2(t,\lambda_t)\\
&\left.+\rho\frac{\partial^2 F}{\partial x\partial y}(t,r_t,\lambda_t)\sigma_1(t,r_t)\sigma_2(t,\lambda_t)+\frac{\partial F}{\partial t}(t,r_t,\lambda_t)\right)dt,
\end{aligned}
$$

hence the bond pricing PDE is

$$
-(x+y)F(t,x,y)+\mu_1(t,x)\frac{\partial F}{\partial x}(t,x,y)
$$

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$$+\mu_2(t,y)\frac{\partial F}{\partial y}(t,x,y)+\frac{1}{2}\sigma_1^2(t,x)\frac{\partial^2 F}{\partial x^2}(t,x,y)$$
$$+\frac{1}{2}\sigma_2^2(t,y)\frac{\partial^2 F}{\partial y^2}(t,x,y)+\rho\sigma_1(t,x)\sigma_2(t,y)\frac{\partial^2 F}{\partial x\partial y}(t,x,y)+\frac{\partial F}{\partial t}(t,r_t,\lambda_t)=0.$$

d) We have

$$\begin{aligned}P(t,T)&=\mathbb{1}_{\{\tau>t\}}\mathbb{E}\left[\exp\left(-\int_t^T r_s ds-\int_t^T\lambda_s ds\right)\,\middle|\,\mathcal{F}_t\right]\\&=\mathbb{1}_{\{\tau>t\}}\exp\left(-\mathbb{E}\left[\int_t^T r_s ds\,\middle|\,\mathcal{F}_t\right]-\mathbb{E}\left[\int_t^T\lambda_s ds\,\middle|\,\mathcal{F}_t\right]\right)\\&\quad\times\exp\left(\frac{1}{2}\,\mathrm{Var}\left[\int_t^T r_s ds+\int_t^T\lambda_s ds\,\middle|\,\mathcal{F}_t\right]\right)\\&=\mathbb{1}_{\{\tau>t\}}\exp\left(-C(a,t,T)r_t-C(b,t,T)\lambda_t\right)\\&\quad\times\exp\left(\frac{\sigma^2}{2}\int_t^T C^2(a,s,T)ds+\frac{\eta^2}{2}\int_t^T C^2(b,s,T)\mathrm{e}^{-(T-s)b}ds\right)\\&\quad\times\exp\left(\rho\sigma\eta\int_t^T C(a,s,T)C(b,s,T)ds\right).\end{aligned}$$

e) This is a direct consequence of the answers to Questions (c) and (d).

f) The above analysis shows that

$$\begin{aligned}\mathbb{P}(\tau>T\mid\mathcal{G}_t)&=\mathbb{1}_{\{\tau>t\}}\mathbb{E}\left[\exp\left(-\int_t^T\lambda_s ds\right)\,\middle|\,\mathcal{F}_t\right]\\&=\mathbb{1}_{\{\tau>t\}}\exp\left(-C(b,t,T)\lambda_t+\frac{\eta^2}{2}\int_t^T C^2(b,s,T)ds\right),\end{aligned}$$

for $a=0$ and

$$\mathbb{E}\left[\exp\left(-\int_t^T r_s ds\right)\,\middle|\,\mathcal{F}_t\right]=\exp\left(-C(a,t,T)r_t+\frac{\sigma^2}{2}\int_t^T C^2(a,s,T)ds\right),$$

for $b=0$, and this implies

$$\begin{aligned}U_\rho(t,T)&=\exp\left(\rho\sigma\eta\int_t^T C(a,s,T)C(b,s,T)ds\right)\\&=\exp\left(\rho\frac{\sigma\eta}{ab}\left(T-t-C(a,t,T)-C(b,t,T)+C(a+b,t,T)\right)\right).\end{aligned}$$

g) We have

$$f(t,T)=-\mathbb{1}_{\{\tau>t\}}\frac{\partial}{\partial T}\log P(t,T)$$

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$$= \mathbb{1}_{\{\tau>t\}} \left(r_t \mathrm{e}^{-(T-t)a} - \frac{\sigma^2}{2} C^2(a,t,T) + \lambda_t \mathrm{e}^{-(T-t)b} - \frac{\eta^2}{2} C^2(b,t,T) \right)$$
$$- \mathbb{1}_{\{\tau>t\}} \rho\sigma\eta C(a,t,T) C(b,t,T).$$

h) We use the relation

$$\begin{aligned}
\mathbb{P}(\tau > T \mid \mathcal{G}_t) &= \mathbb{1}_{\{\tau>t\}} \mathbb{E}\left[\exp\left(-\int_t^T \lambda_s ds \right) \,\middle|\, \mathcal{F}_t \right] \\
&= \mathbb{1}_{\{\tau>t\}} \exp\left(-C(b,t,T)\lambda_t + \frac{\eta^2}{2} \int_t^T C^2(b,s,T) ds \right) \\
&= \mathbb{1}_{\{\tau>t\}} \mathrm{e}^{-\int_t^T f_2(t,u) du},
\end{aligned}$$

where $f_2(t,T)$ is the Vasicek forward rate corresponding to λ_t, *i.e.*

$$f_2(t,u) = \lambda_t \mathrm{e}^{-(u-t)b} - \frac{\eta^2}{2} C^2(b,t,u).$$

i) In this case we have $\rho = 0$ and

$$P(t,T) = \mathbb{P}(\tau > T \mid \mathcal{G}_t) \mathbb{E}\left[\exp\left(-\int_t^T r_s ds \right) \,\middle|\, \mathcal{F}_t \right],$$

since $U_\rho(t,T) = 0$.

Chapter 10

Exercise 10.1 It suffices to check that as λ tends to ∞, the ratio

$$\frac{S_{T_i}^{i,j} \sum_{k=i}^{j-1} \delta_k \exp\left(-\sum_{p=i}^{k} \delta_p r_p - \lambda \sum_{p=i}^{k} \delta_p \right)}{(1-\xi) \sum_{k=i}^{j-1} \left(1 - \mathrm{e}^{-\lambda\delta_k}\right) \exp\left(-\sum_{p=i}^{k} \delta_p r_p - \lambda \sum_{p=i}^{k} \delta_p \right)}$$

converges to 0, while it tends to $+\infty$ as λ goes to 0. Therefore, the equation (10.5) admits a numerical solution λ_0 by continuity of the above function in the variable λ.

Exercise 10.2 Equation (10.5) reads

$$S_{T_i}^{i,j}\sum_{k=i}^{j-1}\delta_k\exp\left(-\sum_{p=i}^{k}\delta_p r_p-\lambda\sum_{p=i}^{k}\delta_p\right)$$
$$=(1-\xi)\sum_{k=i}^{j-1}\left(\mathrm{e}^{\lambda\delta_k}-1\right)\exp\left(-\sum_{p=i}^{k}\delta_p r_p-\lambda\sum_{p=i}^{k}\delta_p\right),$$

or

$$S_{T_i}^{i,j}\sum_{k=i}^{j-1}\delta_k P(0,T_{k+1})\exp\left(-\lambda\sum_{p=i}^{k}\delta_p\right)$$
$$=(1-\xi)\sum_{k=i}^{j-1}\left(\mathrm{e}^{\lambda\delta_k}-1\right)P(0,T_{k+1})\exp\left(-\lambda\sum_{p=i}^{k}\delta_p\right),$$

since

$$P(0,T_{k+1})=\exp\left(-\sum_{p=i}^{k}\delta_p r_p\right),\quad k=0,1,2.$$

From the terminal data of Figure 10.6 we infer that

$$\begin{cases} i=0,\ j=3,\\ t=04/12/2015,\\ T_0=03/22/2015,\ T_1=06/22/2015,\ T_2=09/22/2015,\ T_3=12/22/2015,\\ \delta_1=T_1-t=71/365,\ \delta_2=T_2-T_1=91/365,\ \delta_3=91/365,\\ \xi=0.4,\ S_t^{0,3}=0.001079,\\ P(t,T_1)=0.99952277, P(t,T_2)=0.99827639,\ P(t,T_3)=0.99607821. \end{cases}$$

Hence, from the data of Figure 10.7, Equation (10.5) rewrites as

$$\begin{aligned} &S_t^{0,3}\big(\delta_1 P(t,T_1)\mathrm{e}^{-\lambda\delta_1}+\delta_2 P(t,T_2)\mathrm{e}^{-\lambda\delta_2}+\delta_2 P(t,T_3)\mathrm{e}^{-\lambda\delta_2}\big)\\ &=(1-\xi)\big(P(t,T_1)\times\big(1-\mathrm{e}^{-(T_1-t)\lambda}\big)+P(t,T_2)\times\big(\mathrm{e}^{-(T_2-t)\lambda}-\mathrm{e}^{-(T_3-t)\lambda}\big)\\ &\quad+P(t,T_3)\times\big(\mathrm{e}^{-(T_3-t)\lambda}-\mathrm{e}^{-(T_2-t)\lambda}\big)\big), \end{aligned}$$

or, using the discount factors of Figure 10.7 in Table S.4,

$$0.001079\left(0.99952277\times\mathrm{e}^{-\lambda\times 71/365}\times\frac{71}{365}+0.99827639\times\mathrm{e}^{-\lambda\times 162/365}\times\frac{91}{365}\right.$$
$$\left.+0.99607821\times\mathrm{e}^{-\lambda\times 253/365}\times\frac{91}{365}\right)\tag{A.4}$$

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$$= (1-0.4)\times$$
$$\times \left(0.99952277 \times \left(1-\mathrm{e}^{-\lambda\times 71/365}\right) + 0.99827639 \times \left(\mathrm{e}^{-\lambda\times 71/365} - \mathrm{e}^{-\lambda\times 162/365}\right)\right.$$
$$\left. + 0.99607821 \times \left(\mathrm{e}^{-\lambda\times 162/365} - \mathrm{e}^{-\lambda\times 253/365}\right)\right).$$

Next, we solve Equation (A.4) numerically to find the default rate $\lambda_1 = 0.0017987468$. The default probability is given by

$$p = 1 - \mathrm{e}^{-\lambda\times 253/365} = 0.0012460256,$$

which is consistent with the value of 0.0013 in Figure 10.6, see also Castellacci (2008).

Date	Delta	Discount Factor	Premium Leg	Protection Leg
Jun 22, 2015	0.2611111	0.99952277	0.0002814722	0.0002814708
Sep 22, 2015	0.2527778	0.99827639	0.0002721533	0.000272154
Dec 22, 2015	0.2527778	0.99607821	0.0002715541	0.0002715548
		Sum	0.0008251796	0.0008251796

Table S.4: CDS Market data.

```
P1=0.99952277; P2=0.99827639; P3=0.99607821; S=0.001079;xi=0.4
T0=78/365; t=101/365; T1=172/365; T2=263/365; T3=354/365
delta0=T1-t; delta1=T2-T1; delta2=T3-T2
f <- function(x){return (
S * (delta0 * P1 * exp(-x*(T1-t)) + delta1 * P2 * exp(-x*(T2-t)) + delta2 * P3 * exp(-x*(T3-t)))
- (1 - xi) * (P1 * ( 1 - exp(-x*(T1-t)))
+ P2 * ( exp(-x*(T1-t)) - exp(-x*(T2-t)))
+ P3 * ( exp(-x*(T2-t)) - exp(-x*(T3-t)))))}
x=seq(0,0.004,0.0001); par(mar = c(5, 6, 4, 2))
plot(x,f(x),col='blue',lwd=2,type='l',xaxs='i', cex.lab=1.6, cex.axis=1.4)
abline(h=0); abline(v=0.0017987468); 1-exp(-0.0017987468*(T3-t))
```

Listing S.13: R code - CDS Pricing.

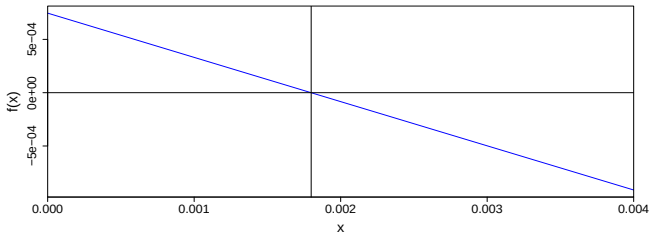


Fig. S.15: Failure rate estimation.

Exercise 10.3

a) We have

$$\begin{aligned}
&\sum_{k=i}^{j-1}\mathbb{E}\left[\mathbb{1}_{(T_k,T_{k+1}]}(\tau)(1-\xi_{k+1})\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\ \Big|\ \mathcal{G}_t\right]\\
&=\sum_{k=i}^{j-1}\mathbb{E}\left[(\mathbb{1}_{\{T_k<\tau\}}-\mathbb{1}_{\{T_{k+1}<\tau\}})(1-\xi_{k+1})\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\ \Big|\ \mathcal{G}_t\right]\\
&=\mathbb{1}_{\{\tau>t\}}\sum_{k=i}^{j-1}\mathbb{E}\left[(1-\xi_{k+1})\left(\mathrm{e}^{-\int_t^{T_k}\lambda_s ds}-\mathrm{e}^{-\int_t^{T_{k+1}}\lambda_s ds}\right)\mathrm{e}^{-\int_t^{T_{k+1}}r(s)ds}\ \Big|\ \mathcal{F}_t\right]\\
&=\mathbb{1}_{\{\tau>t\}}(1-\xi)\sum_{k=i}^{j-1}\mathrm{e}^{-\int_t^{T_{k+1}}r(s)ds}\mathbb{E}\left[\mathrm{e}^{-\int_t^{T_k}\lambda_s ds}-\mathrm{e}^{-\int_t^{T_{k+1}}\lambda_s ds}\ \Big|\ \mathcal{F}_t\right]\\
&=\mathbb{1}_{\{\tau>t\}}(1-\xi)\sum_{k=i}^{j-1}P(t,T_{k+1})\left(Q(t,T_k)-Q(t,T_{k+1})\right).
\end{aligned}$$

b) We have

$$\begin{aligned}
V^p(t,T)&=S_t^{i,j}\sum_{k=i}^{j-1}\delta_k\mathbb{E}\left[\mathbb{1}_{\{\tau>T_{k+1}\}}\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\ \Big|\ \mathcal{G}_t\right]\\
&=S_t^{i,j}\sum_{k=i}^{j-1}\delta_k\mathbb{E}\left[\mathbb{1}_{\{T_{k+1}<\tau\}}\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\ \Big|\ \mathcal{G}_t\right]\\
&=\mathbb{1}_{\{\tau>t\}}S_t^{i,j}\sum_{k=i}^{j-1}\delta_k\mathbb{E}\left[\exp\left(-\int_t^{T_{k+1}}\lambda_s ds\right)\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\ \Big|\ \mathcal{F}_t\right]\\
&=S_t^{i,j}\mathbb{1}_{\{\tau>t\}}\sum_{k=i}^{j-1}\delta_k\exp\left(-\int_t^{T_{k+1}}r(s)ds\right)\mathbb{E}\left[\exp\left(-\int_t^{T_{k+1}}\lambda_s ds\right)\ \Big|\ \mathcal{F}_t\right]\\
&=\mathbb{1}_{\{\tau>t\}}S_t^{i,j}\sum_{k=i}^{j-1}\delta_k P(t,T_{k+1})Q(t,T_{k+1}).
\end{aligned}$$

c) By equating the protection and premium legs, we find

$$\begin{aligned}
&(1-\xi)\sum_{k=i}^{j-1}P(t,T_{k+1})\left(Q(t,T_k)-Q(t,T_{k+1})\right)\\
&\qquad=S_t^{i,j}\sum_{k=i}^{j-1}\delta_k P(t,T_{k+1})Q(t,T_{k+1}).
\end{aligned}$$

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For $j = i+1$, this yields

$$(1-\xi)P(t,T_{i+1})\left(Q(t,T_i)-Q(t,T_{i+1})\right) = S_t^{i,i+1}\delta_i P(t,T_{i+1})Q(t,T_{i+1}),$$

hence

$$Q(t,T_{i+1}) = \frac{1-\xi}{S_t^{i,i+1}\delta_i + 1 - \xi},$$

with $Q(t,T_i)=1$, and the recurrence relation

$$\begin{aligned}
&(1-\xi)P(t,T_{j+1})\left(Q(t,T_j)-Q(t,T_{j+1})\right)\\
&\quad + (1-\xi)\sum_{k=i}^{j-1} P(t,T_{k+1})\left(Q(t,T_k)-Q(t,T_{k+1})\right)\\
&\quad = S_t^{i,j}\delta_j P(t,T_{j+1})Q(t,T_{j+1}) + S_t^{i,j}\sum_{k=i}^{j-1}\delta_k P(t,T_{k+1})Q(t,T_{k+1}),
\end{aligned}$$

i.e.

$$\begin{aligned}
Q(t,T_{j+1}) &= \frac{(1-\xi)Q(t,T_j)}{1-\xi+S_t^{i,j}\delta_j} \qquad (A.5)\\
&\quad + \sum_{k=i}^{j-1}\frac{P(t,T_{k+1})\big((1-\xi)Q(t,T_k)-Q(t,T_{k+1})\big((1-\xi)+\delta_k S_t^{i,j}\big)\big)}{P(t,T_{j+1})(1-\xi+S_t^{i,j}\delta_j)}.
\end{aligned}$$

Exercise 10.4 (Exercise 10.3 continued). Using the recurrence relation (A.5) and the terminal data of Figure 10.8, we find the spread data and survival probabilities in Table S.5.

k	Maturity	T_k	$S_t^{1,k}$ (bp)	$Q(t,T_k)$
1	6M	0.5	10.97	0.999087
2	1Y	1	12.25	0.997961
3	2Y	2	14.32	0.995235
4	3Y	3	19.91	0.990037
5	4Y	4	26.48	0.982293
6	5Y	5	33.29	0.972122
7	7Y	7	52.91	0.937632
8	10Y	10	71.91	0.880602

Table S.5: Spread and survival probabilities.

Chapter 11

Exercise 11.1

a) We have

$$\mathbb{E}[X \mid G] = \lambda_G \int_0^\infty x \mathrm{e}^{-\lambda_G x} dx = \frac{1}{\lambda_G}$$

and

$$\mathbb{E}[X \mid B] = \lambda_B \int_0^\infty x \mathrm{e}^{-\lambda_B x} dx = \frac{1}{\lambda_B}.$$

b) We find

$$\begin{aligned}
\mathbb{P}(B \mid X = x) &= \frac{f_X(x \mid B)\mathbb{P}(B)}{f_X(x \mid G)\mathbb{P}(G) + f_X(x \mid B)\mathbb{P}(B)} \\
&= \frac{\lambda_B \mathrm{e}^{-\lambda_B x}\mathbb{P}(B)}{\lambda_G \mathrm{e}^{-\lambda_G x}\mathbb{P}(G) + \lambda_B \mathrm{e}^{-\lambda_B x}\mathbb{P}(B)} \\
&= \frac{1}{1 + \dfrac{\lambda_G \mathbb{P}(G)}{\lambda_B \mathbb{P}(B)} \mathrm{e}^{(\lambda_B - \lambda_G)x}} \\
&= \frac{1}{1 + \lambda(x)\dfrac{\mathbb{P}(G)}{\mathbb{P}(B)}},
\end{aligned}$$

where $\lambda(x)$ is the *likelihood ratio*

$$\lambda(x) = \frac{f_X(x \mid G)}{f_X(x \mid B)} = \frac{\lambda_G}{\lambda_B} \mathrm{e}^{(\lambda_B - \lambda_G)x}, \qquad x > 0.$$

c) The condition

$$D\mathbb{P}(B \mid X = x) \leqslant L\mathbb{P}(G \mid X = x)$$

rewrites as

$$D\mathbb{P}(B \mid X = x) \leqslant L(1 - \mathbb{P}(B \mid X = x)),$$

i.e.

$$(L + D)\mathbb{P}(B \mid X = x) \leqslant L$$

or

$$\frac{L + D}{1 + \lambda(x)\dfrac{\mathbb{P}(G)}{\mathbb{P}(B)}} \leqslant L,$$

or

$$\lambda(x) = \frac{\lambda_G}{\lambda_B} \mathrm{e}^{(\lambda_B - \lambda_G)x} \geqslant \frac{D}{L} \frac{\mathbb{P}(B)}{\mathbb{P}(G)}.$$

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This condition holds if and only if

$$x \geqslant \frac{1}{\lambda_B - \lambda_G} \log\left(\frac{D}{L}\frac{\lambda_B \mathbb{P}(B)}{\lambda_G \mathbb{P}(G)}\right),$$

provided that $\lambda_B > \lambda_G$. Therefore, we have

$$\begin{aligned}\mathscr{A} &= \left\{x \in \mathbb{R} \ : \ \lambda(x) \geqslant \frac{D}{L}\frac{\mathbb{P}(B)}{\mathbb{P}(G)}\right\} \\ &= \left[\frac{1}{\lambda_B - \lambda_G} \log\left(\frac{D}{L}\frac{\lambda_B \mathbb{P}(B)}{\lambda_G \mathbb{P}(G)}\right), \infty\right),\end{aligned}$$

under the condition

$$\mathbb{E}[X \mid B] = \frac{1}{\lambda_B} < \frac{1}{\lambda_G} = \mathbb{E}[X \mid G].$$

Exercise 11.2

a) We find

$$\begin{aligned}\mathbb{P}(B \mid X = x) &= \frac{f_X(x \mid B)\mathbb{P}(B)}{f_X(x \mid G)\mathbb{P}(G) + f_X(x \mid B)\mathbb{P}(B)} \\ &= \mathbb{1}_{[0,\lambda_B]}(x)\frac{\mathbb{P}(B)/\lambda_B}{\mathbb{P}(G)/\lambda_G + \mathbb{P}(B)/\lambda_B} \\ &= \mathbb{1}_{[0,\lambda_B]}(x)\frac{1}{1 + \dfrac{\lambda_B \mathbb{P}(G)}{\lambda_G \mathbb{P}(B)}}.\end{aligned}$$

b) We have

$$\mathbb{E}[X \mid G] = \int_{-\infty}^{\infty} y f_X(y \mid G)dy = \frac{1}{\lambda_G}\int_0^{\lambda_G} y dy = \frac{\lambda_G}{2},$$

and similarly

$$\mathbb{E}[X \mid B] = \int_{-\infty}^{\infty} y f_X(y \mid B)dy = \frac{1}{\lambda_B}\int_0^{\lambda_B} y dy = \frac{\lambda_B}{2}.$$

c) The condition

$$D\mathbb{P}(B \mid X = x) \leqslant L\mathbb{P}(G \mid X = x)$$

rewrites as

$$D\mathbb{P}(B \mid X = x) \leqslant L(1 - \mathbb{P}(B \mid X = x)),$$

i.e.

$$(L+D)\mathbb{P}(B \mid X=x) \leqslant L$$

or

$$D\mathbb{1}_{[0,\lambda_B]}(x) \leqslant L\mathbb{1}_{(\lambda_B,\infty)}(x) + L\frac{\lambda_B \mathbb{P}(G)}{\lambda_G \mathbb{P}(B)}.$$

This condition holds if and only if

$$\frac{\lambda_B}{\lambda_G} \geqslant \frac{D}{L}\frac{\mathbb{P}(B)}{\mathbb{P}(G)}.$$

when $x \in [0,\lambda_B]$, and is always satisfied when $x \in (\lambda_B,\infty)$. Therefore, we have $\mathscr{A} = \mathbb{R}$ if

$$\frac{\lambda_B}{\lambda_G} \geqslant \frac{D}{L}\frac{\mathbb{P}(B)}{\mathbb{P}(G)},$$

and $\mathscr{A} = (\lambda_B,\infty)$ if

$$\frac{\lambda_B}{\lambda_G} < \frac{D}{L}\frac{\mathbb{P}(B)}{\mathbb{P}(G)},$$

under the condition

$$\mathbb{E}[X \mid B] = \lambda_B < \lambda_G = \mathbb{E}[X \mid G].$$

Exercise 11.3

a) We have

$$\overline{F}_G(x) = \mathrm{e}^{-\lambda_G x} \quad \text{and} \quad \overline{F}_B(x) = \mathrm{e}^{-\lambda_B x}, \quad x \geqslant 0,$$

hence

$$\overline{F}_B^{-1}(y) = -\frac{\log y}{\lambda_B}, \quad y \in (0,1],$$

and

$$\begin{aligned}\overline{F}_G\big(\overline{F}_B^{-1}(y)\big) &= \overline{F}_G\left(-\frac{\log y}{\lambda_B}\right)\\ &= \mathrm{e}^{\lambda_G(\log y)/\lambda_B}\\ &= y^{\lambda_G/\lambda_B}, \qquad y \in [0,1].\end{aligned}$$

We check that, according to Proposition 11.9,

$$\begin{aligned}\frac{d}{dy}\overline{F}_G\big(\overline{F}_B^{-1}(y)\big) &= \frac{d}{dy}y^{\lambda_G/\lambda_B}\\ &= \frac{d}{dy}\mathrm{e}^{\lambda_G(\log y)/\lambda_B}\end{aligned}$$

$$
\begin{aligned}
&= \frac{\lambda_G}{y\lambda_B} \mathrm{e}^{\lambda_G(\log y)/\lambda_B} \\
&= \frac{\lambda_G}{\lambda_B} \mathrm{e}^{(\lambda_G-\lambda_B)(\log y)/\lambda_B} \\
&= \frac{\lambda_G}{\lambda_B} \mathrm{e}^{(\lambda_B-\lambda_G)\overline{F}_B^{-1}(y)} \\
&= \lambda\big(\overline{F}_B^{-1}(y)\big), \qquad x \in [0,1].
\end{aligned}
$$

Figure S.16 presents three samples of exponential ROC curves, with successively $(\lambda_B, \lambda_G) = (10, 1)$, $(\lambda_B, \lambda_G) = (2, 1)$, and $(\lambda_B, \lambda_G) = (1, 1)$.

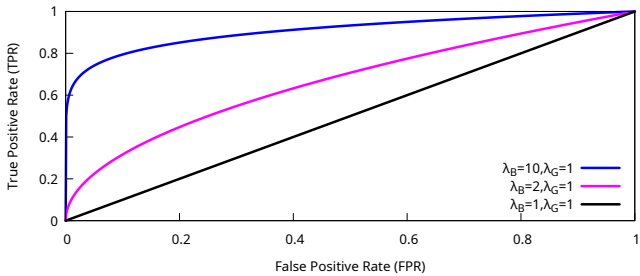


Fig. S.16: Exponential ROC curves.

b) We have

$$\overline{F}_G(x) := 1 - \frac{x}{\lambda_G}, \qquad x \in [0, \lambda_G],$$

and

$$\overline{F}_B(x) := 1 - \frac{x}{\lambda_B}, \qquad x \in [0, \lambda_B],$$

hence

$$\overline{F}_B^{-1}(y) := \lambda_B(1-y), \qquad y \in [0,1],$$

hence

$$\overline{F}_G\big(\overline{F}_B^{-1}(x)\big) = 1 - \frac{\lambda_B}{\lambda_G}(1-y) = \frac{\lambda_G - \lambda_B}{\lambda_G} + \frac{\lambda_B}{\lambda_G} y, \quad y \in [0,1].$$

Figure S.17 presents three samples of uniform ROC curves, with successively $(\lambda_B, \lambda_G) = (1, 8)$, $(\lambda_B, \lambda_G) = (1, 2)$, and $(\lambda_B, \lambda_G) = (1, 1)$.

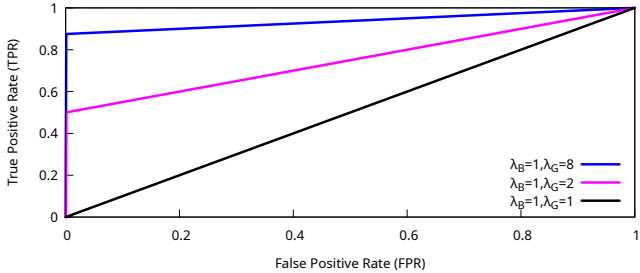


Fig. S.17: Uniform ROC curves.

Exercise 11.4

a) We have

$$\begin{aligned}\mathbb{P}(B \mid X=x) &= \frac{\mathbb{P}(B) f_X(x \mid B)}{\mathbb{P}(G) f_X(x \mid G) + \mathbb{P}(B) f_X(x \mid B)} \\ &= \frac{\mathbb{P}(B) \mathrm{e}^{-(x-\mu_B)^2/(2\sigma^2)}}{\mathbb{P}(G) \mathrm{e}^{-(x-\mu_G)^2/(2\sigma^2)} + \mathbb{P}(B) \mathrm{e}^{-(x-\mu_B)^2/(2\sigma^2)}} \\ &= \frac{1}{1+\mathrm{e}^{\alpha+\beta x}}, \qquad x \in \mathbb{R},\end{aligned}$$

with

$$\beta := \frac{\mu_G - \mu_B}{\sigma^2} > 0$$

and

$$\alpha := -\beta \frac{\mu_G + \mu_B}{2} + \log\left(\frac{\mathbb{P}(G)}{\mathbb{P}(B)}\right).$$

b) We have

$$\begin{aligned}\lambda(x) &= \frac{f_X(x \mid G)}{f_X(x \mid B)} \\ &= \mathrm{e}^{-(x-\mu_G)^2/(2\sigma^2)+(x-\mu_B)^2/(2\sigma^2)} \\ &= \mathrm{e}^{-(\mu_G^2-\mu_B^2-2x(\mu_G-\mu_B))/(2\sigma^2)} \\ &= \mathrm{e}^{\beta x-(\mu_G^2-\mu_B^2)/(2\sigma^2)}, \qquad x \in \mathbb{R}.\end{aligned}$$

c) The condition

$$\lambda(x) = \mathrm{e}^{\beta x - (\mu_G^2 - \mu_B^2)/(2\sigma^2)} \geqslant \frac{D(x)}{L(x)} \frac{\mathbb{P}(B)}{\mathbb{P}(G)}$$

is equivalent to

$$\begin{aligned}\beta x &\geqslant \frac{\mu_G^2 - \mu_B^2}{2\sigma^2} + \log\left(\frac{D(x)}{L(x)} \frac{\mathbb{P}(B)}{\mathbb{P}(G)}\right) \\ &\geqslant \frac{\mu_G^2 - \mu_B^2}{2\sigma^2} + \log \frac{\mathbb{P}(B)}{\mathbb{P}(G)} + \log \frac{D(x)}{L(x)} \\ &\geqslant \frac{\mu_G^2 - \mu_B^2}{2\sigma^2} + \log \frac{\mathbb{P}(B)}{\mathbb{P}(G)} + x(a+b),\end{aligned}$$

hence

$$x \geqslant \frac{\mu_G^2 - \mu_B^2}{2\sigma^2(\beta - a - b)} + \frac{1}{\beta - a - b} \log \frac{\mathbb{P}(B)}{\mathbb{P}(G)},$$

provided that

$$\beta := \frac{\mu_G - \mu_B}{\sigma^2} > a + b.$$

In this case, we have

$$\mathscr{A}^* = [x^*, \infty) = \left[\frac{\mu_G^2 - \mu_B^2}{2\sigma^2(\beta - a - b)} + \frac{1}{\beta - a - b} \log \frac{\mathbb{P}(B)}{\mathbb{P}(G)}, \infty\right),$$

where

$$x^* := \frac{\mu_G^2 - \mu_B^2}{2\sigma^2(\beta - a - b)} + \frac{1}{\beta - a - b} \log \frac{\mathbb{P}(B)}{\mathbb{P}(G)}.$$

Background on Probability Theory

Exercise A.1

a) We have

$$\begin{aligned}\mathbb{E}[X] &= \sum_{k \geqslant 0} k \mathbb{P}(X = k) = \mathrm{e}^{-\lambda} \sum_{k \geqslant 0} k \frac{\lambda^k}{k!} \\ &= \mathrm{e}^{-\lambda} \sum_{k \geqslant 1} \frac{\lambda^k}{(k-1)!} = \lambda \mathrm{e}^{-\lambda} \sum_{k \geqslant 0} \frac{\lambda^k}{k!} = \lambda.\end{aligned}$$

b) We have

$$\begin{aligned}\mathbb{E}[X^2] &= \sum_{k\geqslant 0} k^2 \mathbb{P}(X=k)\\ &= \mathrm{e}^{-\lambda}\sum_{k\geqslant 1} k^2\frac{\lambda^k}{k!}\\ &= \mathrm{e}^{-\lambda}\sum_{k\geqslant 1} k\frac{\lambda^k}{(k-1)!}\\ &= \mathrm{e}^{-\lambda}\sum_{k\geqslant 2}\frac{\lambda^k}{(k-2)!} + \mathrm{e}^{-\lambda}\sum_{k\geqslant 1}\frac{\lambda^k}{(k-1)!}\\ &= \lambda^2\mathrm{e}^{-\lambda}\sum_{k\geqslant 0}\frac{\lambda^k}{k!} + \lambda\mathrm{e}^{-\lambda}\sum_{k\geqslant 0}\frac{\lambda^k}{k!}\\ &= \lambda^2+\lambda,\end{aligned}$$

and

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda = \mathbb{E}[X].$$

Exercise A.2 We have

$$\begin{aligned}\mathbb{P}(\mathrm{e}^X > c) = \mathbb{P}(X > \log c) &= \int_{\log c}^{\infty} \mathrm{e}^{-y^2/(2\eta^2)}\frac{dy}{\sqrt{2\pi\eta^2}}\\ &= \int_{(\log c)/\eta}^{\infty} \mathrm{e}^{-y^2/2}\frac{dy}{\sqrt{2\pi}} = 1-\Phi((\log c)/\eta) = \Phi(-(\log c)/\eta).\end{aligned}$$

Exercise A.3

a) Using the change of variable $z=(x-\mu)/\sigma$, we have

$$\begin{aligned}\int_{-\infty}^{\infty}\varphi(x)dx &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}dx\\ &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2/(2\sigma^2)}dy\\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\mathrm{e}^{-z^2/2}dz.\end{aligned}$$

Next, using the polar change of coordinates $r^2 = x^2+y^2$ and $dxdy = rdrd\theta$, we find*

* "In a discussion with Grothendieck, Messing mentioned the formula expressing the integral of e^{-x^2} in terms of π, which is proved in every calculus course. Not only did Grothendieck not know the formula, but he thought that he had never seen it in his life". Milne (2005).

$$\begin{aligned}\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\mathrm{e}^{-z^2/2}dz\right)^2 &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2/2}dy\int_{-\infty}^{\infty}\mathrm{e}^{-z^2/2}dz\\ &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\mathrm{e}^{-(y^2+z^2)/2}dydz\\ &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\infty}r\mathrm{e}^{-r^2/2}drd\theta\\ &= \int_0^{\infty}r\mathrm{e}^{-r^2/2}dr\\ &= \lim_{R\to+\infty}\int_0^{R}r\mathrm{e}^{-r^2/2}dy\\ &= -\lim_{R\to+\infty}\left[\mathrm{e}^{-r^2/2}\right]_0^{R}\\ &= \lim_{R\to+\infty}(1-\mathrm{e}^{-R^2/2})\\ &= 1,\end{aligned}$$

or

$$\int_{-\infty}^{\infty}\mathrm{e}^{-z^2/2}dz=\sqrt{2\pi}.$$

b) We have

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty}x\varphi(x)dx\\ &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}x\mathrm{e}^{-(x-\mu)^2/(2\sigma^2)}dx\\ &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}(\mu+y)\mathrm{e}^{-y^2/(2\sigma^2)}dx\\ &= \frac{\mu}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2/2}dy+\frac{\sigma}{\sqrt{2\pi}}\int_{-\infty}^{\infty}y\mathrm{e}^{-y^2/2}dy\\ &= \frac{\mu}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2/2}dy+\frac{\sigma}{\sqrt{2\pi}}\lim_{A\to+\infty}\int_{-A}^{A}y\mathrm{e}^{-y^2/2}dy\\ &= \frac{\mu}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\mathrm{e}^{-y^2/2}dy\\ &= \mu\int_{-\infty}^{\infty}\varphi(y)dy\\ &= \mu\mathbb{P}(X\in\mathbb{R})\\ &= \mu,\end{aligned}$$

by symmetry of the function $y\longmapsto y\mathrm{e}^{-y^2/2}$ on $\mathbb{R}$.

c) Similarly, by integration by parts twice on $\mathbb{R}$, we find

$$\mathbb{E}[(X-\mathbb{E}[X])^2]=\int_{-\infty}^{\infty}(x-\mu)^2\varphi(x)dx$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} y^2 \mathrm{e}^{-(y-\mu)^2/(2\sigma^2)} dy \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y \times y \mathrm{e}^{-y^2/2} dy \\
&= \frac{\sigma^2}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{-y^2/2} dy \\
&= \sigma^2.
\end{aligned}
$$

d) By a completion of squares argument, we have

$$
\begin{aligned}
\mathbb{E}[\mathrm{e}^X] &= \int_{-\infty}^{\infty} \mathrm{e}^x \varphi(x) dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \mathrm{e}^{x-(x-\mu)^2/(2\sigma^2)} dx \\
&= \frac{\mathrm{e}^\mu}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \mathrm{e}^{y-y^2/(2\sigma^2)} dy \\
&= \frac{\mathrm{e}^\mu}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \mathrm{e}^{\sigma^2/2+(y-\sigma^2)^2/(2\sigma^2)} dy \\
&= \frac{\mathrm{e}^\mu + \sigma^2/2}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} \mathrm{e}^{x^2/(2\sigma^2)} dy \\
&= \mathrm{e}^\mu + \frac{\sigma^2}{2}.
\end{aligned}
$$

Exercise A.4

a) We have

$$
\begin{aligned}
\mathbb{E}[X^+] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x^+ \mathrm{e}^{-x^2/(2/\sigma^2)} dx \\
&= \frac{\sigma}{\sqrt{2\pi}} \int_0^{\infty} x \mathrm{e}^{-x^2/2} dx \\
&= \frac{\sigma}{\sqrt{2\pi}} \left[-\mathrm{e}^{-x^2/2} \right]_{x=0}^{x=\infty} \\
&= \frac{\sigma}{\sqrt{2\pi}}.
\end{aligned}
$$

b) We have

$$
\begin{aligned}
\mathbb{E}[(X-K)^+] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} (x-K)^+ \mathrm{e}^{-x^2/(2\sigma^2)} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_K^{\infty} (x-K) \mathrm{e}^{-x^2/(2\sigma^2)} dx
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_K^\infty x\mathrm{e}^{-x^2/(2\sigma^2)} dx - \frac{K}{\sqrt{2\pi\sigma^2}} \int_K^\infty \mathrm{e}^{-x^2/(2\sigma^2)} dx \\
&= \frac{\sigma}{\sqrt{2\pi}} \left[-\mathrm{e}^{-x^2/(2\sigma^2)} \right]_{x=K}^\infty - \frac{K}{\sqrt{2\pi}} \int_{-\infty}^{-K/\sigma} \mathrm{e}^{-x^2/2} dx \\
&= \frac{\sigma}{\sqrt{2\pi}} \mathrm{e}^{-K^2/(2\sigma)^2} - K\Phi\left(-\frac{K}{\sigma}\right).
\end{aligned}
$$

c) Similarly, we have

$$
\begin{aligned}
\mathbb{E}[(K-X)^+] &= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^\infty (K-x)^+ \mathrm{e}^{-x^2/(2\sigma^2)} dx \\
&= \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^K (K-x) \mathrm{e}^{-x^2/(2\sigma^2)} dx \\
&= \frac{K}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^K \mathrm{e}^{-x^2/(2\sigma^2)} dx - \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^K x\mathrm{e}^{-x^2/(2\sigma^2)} dx \\
&= \frac{K}{\sqrt{2\pi}} \int_{-\infty}^{K/\sigma} \mathrm{e}^{-x^2/2} dx - \frac{\sigma}{\sqrt{2\pi}} \left[-\mathrm{e}^{-x^2/(2\sigma^2)} \right]_{-\infty}^{x=K} \\
&= \frac{\sigma}{\sqrt{2\pi}} \mathrm{e}^{-K^2/(2\sigma)^2} + K\Phi\left(\frac{K}{\sigma}\right).
\end{aligned}
$$